

Public Transport Electrification: Lessons from a Pioneer*

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Abstract

Cities worldwide are committing to zero-emission bus fleets. We evaluate a pioneer: Santiago, Chile, whose electric bus fleet is the largest outside China. We link the universe of georeferenced bus trips to vehicle technology, ridership, and nearby air and noise monitors. We exploit staggered re-tenders triggered by contract expirations set before electrification entered the policy agenda. Fleet electrification improves local air quality and reduces noise near routes, with little change in operating speed and no detectable ridership gain. At the scale reached in 2025, these environmental gains imply gross annual benefits of USD 128–362 million.

Keywords: electric buses; public transport; air pollution; noise; procurement.

JEL codes: H23, H57, O33, Q53, R41, R48.

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1 Introduction

Climate change has moved the transition to zero-emission technologies to the center of public policy, and a growing literature evaluates the welfare effects of the taxes and subsidies deployed to accelerate it (Acemoglu et al., 2016; Feger et al., 2022; Hahn et al., 2026). The case for this transition rests only partly on carbon: burning fossil fuels also degrades local air quality, and quasi-experimental evidence shows that even short-lived increases in exposure raise mortality, medical spending, and hospital admissions (Schlenker and Walker, 2016; Deryugina et al., 2019). Public buses have become a leading margin of adjustment in this transition, and commitments now span regions across the world. For example, the European Union requires all new urban buses to be zero-emission by 2035, and London has committed to a fully zero-emission fleet by 2034; São Paulo must eliminate fossil CO₂ emissions from its bus fleet by 2038; Jakarta plans to electrify its entire 10,000-bus fleet by 2030; and the largest transit agencies in the United States have committed to fully zero-emission fleets: New York and Chicago by 2040, and Seattle’s King County Metro by 2035.¹

Santiago de Chile offers an unusually good setting to provide that evidence. The city moved early and at scale: its electric bus fleet is the largest of any city outside China, and electric buses account for half of the operating fleet (DTPM, 2025). The setting also offers administrative data of rare quality: the universe of bus trips, georeferenced and linked to ridership; networks of air- and noise-pollution monitors; and microdata on respiratory emergency visits and school absenteeism. Each trip record includes the bus license plate, which we match to the fleet registry identifying every vehicle’s engine technology, so electrification is measured from realized operations, bus by bus and trip by trip, rather than from fleet stocks. Finally, the rollout embeds credible exogenous variation. The timing of electrification across routes was determined by the expiration of concessions signed in 2007 and renegotiated in 2011, years before electrification entered the policy agenda, so deployment timing is unrelated to the pollution burden or the health profile of the areas served (Gómez-Lobo, 2025; Muñoz et al., 2014).

We use these data and this variation to trace the welfare effect of electrification through the margins it can move. Fleet replacement changes the attributes and ridership of bus service, the externalities that bus operations impose on nearby places, and the resource

¹See European Parliament and Council of the EU (2024); Transport for London (2023); Município de São Paulo (2018); The Jakarta Post (2020); Metropolitan Transportation Authority (2024); Chicago Transit Authority (2022); King County Metro Transit (2022) for details.

cost of provision. We use two complementary designs. The staggered start of the new operating contracts identifies the reduced-form effect of the re-tendered service package and provides a predetermined source of variation in local exposure. Separately, for the external outcomes we compare the same location before and after nearby service becomes more electric, holding total bus activity fixed, which isolates the replacement of conventional with electric operation. The first design identifies the bundled contract effect; the second identifies the propulsion-technology replacement effect used in the welfare calculation. These replacement estimates isolate a change in fleet composition rather than a change in the amount of service. The contractual chronology provides the predetermined institutional variation underlying both approaches, while realized deployment additionally reflects operator discretion. Section 3 states this framework and writes the welfare effect as a sum of the components we estimate.

We first document that the new contracts deliver electrification. Because the later tenders required electric vehicles, electric operation should change when an incumbent contract ends and the re-tendered service begins. We observe 11 contract starts between 2022 and 2025, covering 399 treated routes, and compare each affected package of routes with routes that remain outside the new contracts. The electric share of bus-kilometers rises by 12 percentage points in the first complete post-contract week and by a further 3 points per week. Nine weeks after the start it stands 39 points above its pre-contract level, against a pre-contract mean electric share of 16 percent. The change is mostly substitution. Electric bus-kilometers rise by 42 percent of the pre-contract total while conventional kilometers fall by 36 percent, and total bus-kilometers increase by 6 percent. Contract expiration therefore produces large and sharply timed shifts in electric operation. Section 4.2 reports these estimates.

The main results follow this chain. First, the new contracts change how service is supplied, but generate little immediate passenger response. By event week 9, bus-kilometers have increased during peak, off-peak, and weekend periods, while average operating speed changes little and passenger usage and ridership do not rise detectably. In contrast, routes and periods with a higher realized electric share have more buses and bus-kilometers and also exhibit higher usage and ridership. We interpret this difference as evidence that realized deployment contains endogenous operator choices: electric buses may be allocated more intensively to routes or periods where demand is stronger or improving, or alongside other service adjustments. The contract estimates therefore provide the cleaner causal effect of the implemented electrification package, while the realized-share regressions describe how electric deployment and passenger demand move together in equilibrium.

Second, electrification reduced the local externalities of bus travel. Holding total nearby bus activity fixed, replacing conventional with electric bus-kilometers lowers $\text{PM}_{2.5}$, PM_{10} , and NO_2 concentrations, while the estimates for NO_x and CO are close to zero. A mechanical extrapolation to full electrification of the remaining conventional bus activity within 750 meters of a monitoring station implies reductions of roughly 3 percent in $\text{PM}_{2.5}$ and 6 percent in both PM_{10} and NO_2 . As validation, predetermined contract starts—which generated an increase in local electric-bus penetration—are followed by reductions in several pollutants, although these reduced-form estimates also capture broader fleet renewal bundled with the new contracts. Ambient noise also falls by 0.8-0.9 dB within 50-200m after full electrification.

Third, these environmental gains are harder to detect in downstream health and schooling outcomes. Respiratory emergency-room visits decline after nearby service becomes highly electric, but the estimate is imprecise and is accompanied by a similar reduction in non-respiratory visits, limiting a respiratory-specific interpretation. School absenteeism likewise shows no measurable response, both in the realized-electrification design and in the contract-based design, despite a strong contract-induced first stage in local electric-bus exposure. These muted effects are consistent with the magnitude of the pollution reductions we estimate. Mapped through published concentration–response estimates, the implied changes in emergency visits and absences are small relative to the month-to-month variation in these outcomes, while our health and schooling measures are substantially coarser in time and space than the localized daily pollution changes. We therefore treat these outcomes as validation and magnitude checks rather than as independent tests of the pollution channel.

Finally, we value these effects at the official social prices of the Chilean public-investment system and weigh them against costs. On the cost side, the re-tenders electrified the fleet at approximately no additional resource cost: the higher capital cost of electric buses was offset by lower energy and maintenance costs, and the transport authority reports that the 2023 and 2025 tenders reduced annual system payments. Because the incremental resource cost is near zero, the sign and rough magnitude of the welfare effect are set by the externality responses we estimate.

Among these, the mortality gains from lower particulate exposure are the largest benefit, with smaller contributions from less noise and avoided carbon; because the contracts do not raise ridership, no fare-revenue gain enters. Because the mortality valuation drives the total, we report it under two conventions throughout. In 2025, with 42 percent of bus service electric, the benefits amount to USD 362 million per year valuing avoided deaths at the

official statistical-life value, and USD 128 million under a life-years valuation. For scale, the annual public subsidy that finances the system, buses, Metro, and commuter rail combined, is about USD 1.3 billion: the realized gains equal between one-tenth and three-tenths of it. Section 7 reports the full valuation.

Our first contribution is to the economic evidence on green-transition policy. Most of this evidence concerns instruments that operate through private adoption decisions: net energy metering for rooftop solar (Bollinger et al., 2025), public investment in electric-vehicle charging infrastructure (Arkolakis et al., 2025), and the electrification of residential heating (Davis, 2024, 2025). A recurring lesson is that the returns to these instruments hinge on margins the government does not control: how households discount future savings (Bollinger et al., 2025), whether they use the technology at all (Hanna et al., 2016), and the pricing and carbon content of the electricity behind it (Borenstein and Bushnell, 2022; Gillingham et al., 2025). Public bus fleets differ on exactly this margin: the state is the adopter, and the new technology directly replaces diesel buses, among the dirtiest vehicles operating in dense urban corridors. We complement welfare evaluations of green-transition policies (Hahn et al., 2026) with causal evidence on the realized effects of one of the largest state-led electrification programs to date.

Our second contribution is to the literature on technology adoption by the state (Acemoglu et al., 2016). A recent experimental literature studies governments adopting information technologies to improve their own operations: biometric payments and identity verification in welfare programs (Muralidharan et al., 2016, 2025), e-invoicing in workfare (Banerjee et al., 2020), electronic procurement (Lewis-Faupel et al., 2016), electronic tax filing (Okunogbe and Pouliquen, 2022), and automated environmental monitoring (Greenstone et al., 2022; Assunção et al., 2023). These technologies work by generating information that mitigates agency problems inside the state, and their returns hinge on whether the government acts on that information (Dhaliwal and Hanna, 2017; Muralidharan et al., 2025). The technology we study is of a different kind: a production technology embedded in service delivery, whose returns (cleaner air, lower noise) accrue directly to citizens rather than through improved monitoring or enforcement. In Santiago, adoption also ran through competitive procurement, but the object procured was the technology itself. Our evidence documents what this form of state technology adoption delivers, a margin on which systematic evidence remains scarce.

We also contribute to transport economics, and in particular to the literature on policies that reduce the externalities of urban travel. First-best pricing of vehicle use at marginal

external cost (Parry et al., 2007) remains politically and technologically difficult, so cities rely on a menu of second-best instruments: transit subsidies (Parry and Small, 2009; Basso and Silva, 2014), bus priority infrastructure (Basso and Silva, 2014; González and Silva, 2025), lane exemptions that reward cleaner vehicles (Bento et al., 2014), and driving restrictions targeting older vintages, pioneered by Santiago in 1992 (Barahona et al., 2020), with recent work evaluating such policies in general equilibrium (Barwick et al., 2024). Each instrument buys its gains on one margin at a cost on another: subsidies strain public budgets (Parry and Small, 2009), uniform restrictions push households toward additional, dirtier cars (Barahona et al., 2020), and cleaner engines lose potency at congested speeds (Waxman et al., 2024). Judged against this menu, fleet electrification performs unusually well: pollution and noise fall along electrified routes while ridership and operating speeds remain unchanged. Locating a policy with these properties in the second-best toolkit is directly relevant for the growing number of cities committed to zero-emission fleets.

2 Context

Santiago is the capital of Chile and its largest metropolitan area, with a population of about seven million. The city sits in a basin between the coastal mountain range and the Andes, and a persistent subsidence inversion limits vertical mixing. This geography traps local emissions and has made air pollution a first-order policy problem for decades (Barraza et al., 2017). Despite sustained abatement policies since 1990, annual $\text{PM}_{2.5}$ concentrations remain well above the WHO guideline and hover around the Chilean standard (Barraza et al., 2017; IQAir, 2025); panel (c) of Figure 1 places these concentrations in international perspective. Santiago also has one of the world’s longest-running $\text{PM}_{2.5}$ monitoring networks, with continuous records since 1989.

2.1 Transport, pollution, and health

Urban air pollution comes mostly from combustion, and in Santiago transport plays a leading role. The official emissions inventory attributes 20% of the Metropolitan Region’s $\text{PM}_{2.5}$ emissions and 60% of its NO_x emissions to transport (MMA, 2017). Emissions, however, are not the same as exposure: wood burning is concentrated in winter and in peripheral areas, while vehicles emit at street level across the city throughout the year. Source-apportionment estimates based on 15 years of filter samples in central Santiago attribute 37% of ambient

PM_{2.5} to motor vehicles (Barraza et al., 2017).² As panel (d) of Figure 1 shows using data up to 2012, this share was among the highest in the world: it matched the population-weighted average of India and stands well above the world average of 25% (Karagulian et al., 2015).

Within the transport sector, diesel buses are a disproportionate source of harmful local pollutants. They account for about a quarter of the black carbon emitted by the transport sector, and where soot-free engines are not required, a diesel bus can release more than 250 times the black carbon of a gasoline car over the same distance (Climate and Clean Air Coalition, 2017). In much of Latin America the bus fleet still meets only Euro III standards or worse, and until recently only Mexico City and Santiago required Euro VI engines for new buses (Miller et al., 2017; Fun Sang Cepeda et al., 2025). Santiago was no exception before the fleet renewal we study: 82% of its roughly 6,500 buses were Euro III (Climate and Clean Air Coalition, 2017). Moreover, bus emissions are released at street level along dense corridors, so exposure is concentrated where people live, walk, and wait.

A large body of evidence links localized traffic pollution to health. Currie and Walker (2011) find that electronic tolling, by cutting congestion and idling at toll plazas, reduced prematurity by 10.8% and low birth weight by 11.8% among mothers living within 2 km of a plaza. Schlenker and Walker (2016) show that daily variation in airport congestion raises respiratory and heart-related hospital admissions within 10 km of California airports. For Chile, Dardati et al. (2024) estimate that a one-day increase of 1 $\mu\text{g}/\text{m}^3$ in PM_{2.5} raises respiratory emergency-room visits by 0.36%. Replacing diesel buses with zero-emission vehicles can therefore produce health benefits that are large and highly localized, which is the hypothesis we take to the data.

2.2 Bus fleet electrification

Electric buses have become a central tool for decarbonizing urban transport. Their fixed routes, central depots, and shared charging infrastructure make electrification easier to manage than for private vehicles (Avenali et al., 2024). Adoption has accelerated but remains uneven: more than 90% of the world’s roughly 780,000 electric buses operate in

²The full apportionment is 37% motor vehicles, 19% industrial sources, 14% copper smelters, 12% residential wood burning, 10% coastal sources, and 3% urban dust, with wood burning rising to roughly 31% in the cold season (Barraza et al., 2017). On the emissions side, the inventory attributes 39% of PM_{2.5} to residential wood burning, 20% to off-road machinery, and 15% to industry (MMA, 2017).

China, and large fleets outside China remain uncommon (Jaeger, 2025).³ The main obstacle is cost: electric buses carry higher capital costs for batteries and charging, and operators resist adoption without direct public support (Avenali et al., 2023; Giagnorio et al., 2024). Panels (a) and (b) of Figure 1 put Santiago in this global context: it operates the largest electric bus fleet of any city outside China, and close to half of its fleet is electric, among the highest shares outside China (DTPM, 2025).

Santiago’s bus system has been the main testing ground for Chile’s climate commitments, which include an emissions peak in 2025, carbon neutrality by 2050, and, for public transport, zero-emission requirements for all new urban fleet additions by 2035 and fully zero-emission operation by 2040.⁴ In 2019, Santiago became one of the first cities in the world to require Euro VI (or EPA 2010) engines for every new bus, a standard that mandates particulate filters and cuts PM emissions by about 90% relative to Euro V (Climate and Clean Air Coalition, 2017). Electrification started in parallel: the first electric buses arrived as pilots in 2017, and the first large batch of 100 vehicles entered service in late 2018 (DTPM, 2025). Figure 2 documents this transformation: by June 2026, electric buses accounted for 55 percent of the operating fleet (panel a) and roughly two-thirds of daily bus trips (panel b).

2.3 Contracts and rollout

The institutional origins of the rollout date back two decades. Gómez-Lobo (2025) documents how Santiago’s public transport system was overhauled almost overnight on February 10, 2007. This reform, known as Transantiago, simultaneously restructured the route network, the fare collection technology, the bus fleet, and the contracts governing operators. The reform precipitated a combined mobility and political crisis, and the State’s subsequent effort to improve the system cost roughly 0.25% of GDP (Gómez-Lobo, 2025).

The original 2007 concessions assigned markedly different terms to feeder (*alimentador*)

³In Europe, the electric bus fleet grew from about 1,650 in 2018 to more than 9,500 in 2022, and the electric share of new bus sales rose from 2.5% to 10% (European Alternative Fuels Observatory, 2022); a European mandate requires all new urban buses to be zero-emission by 2035 (European Parliament and Council of the EU, 2024).

⁴Chile pledged in 2015 to reduce the emissions intensity of GDP by 30% by 2030 relative to 2007 and, as host of COP25 in 2020, set a maximum emissions budget of 1,100 million tonnes of CO₂-equivalent for 2020–2030. The targets rest on state policies sustained across administrations and led jointly by the Ministries of Energy, Environment, and Transport, including the 2050 Energy Policy, the National Electromobility Strategy, the Energy Efficiency Law, the Energy Storage and Electromobility Law, and the sector’s Sustainable Mobility Policy.

and trunk (*troncal*) services: feeder contracts ran for six years, expiring in 2012, while trunk contracts ran for thirteen years. These concessions were renegotiated during 2011 under the legal framework of Law 20.504 (Muñoz et al., 2014), whose central provision for our purposes governed how the Ministry of Transport could proceed once a concession was terminated early: re-tender, award a three-year direct contract, or award transitional eighteen-month contracts.⁵ Some operators ran to the end of their original term, others accepted three-year direct contracts, and others entered successively renewed eighteen-month arrangements. Contract expirations therefore became staggered across operators and routes. Importantly, both the 2007 terms and the 2011 options were set well before fleet electrification entered the policy agenda, so the expiration schedule was fixed by institutional considerations unrelated to environmental or health objectives.

The link to electrification follows directly. When a contract reached its end, the associated routes were re-tendered, and the later tenders incorporated an electric fleet as a contractual requirement. The arrival of electric buses on a route was thus determined by *when* its incumbent contract happened to expire, rather than by the route’s pollution burden or the health profile of the population it served. A final feature of the re-tenders matters for the welfare analysis. The new contracts separated fleet provision from operation and spread the capital cost of the electric fleet over the contract horizon. This capital premium was approximately offset by the lower energy and maintenance costs of electric operation: the transport authority reports operating-cost savings of 66 percent and maintenance savings of 44 percent for electric buses, and estimates that the 2023 and 2025 re-tenders reduced annual system payments (Directorio de Transporte Público Metropolitano, 2026). The fleet was therefore electrified without a material increase in the resource cost of providing service. We return to this feature in Section 3.

3 Framework and Empirical Design

Fleet electrification replaces one production technology of public transport with another, on a fixed route network and under a fare set citywide by the regulator. Its welfare consequences therefore run through a small number of margins. This section writes the welfare effect of

⁵The Ministry was required to designate a new concessionaire within eighteen months of the termination resolution becoming final, ordinarily through a new public tender. On grounds of public interest and service continuity, it could instead award service directly on a transitional basis, for up to three years or until the end of the original concession term, by decree jointly signed by the Ministers of Transport and Finance.

electrification as a sum of interpretable components, defines the two policy estimands the paper recovers, and distinguishes the empirical designs used to identify them. A central feature of the setting is that electrification is implemented through new operating contracts rather than through isolated drivetrain replacement. The contractual transition therefore provides the cleanest source of causal variation in the transit market, while spatial variation in electric-bus exposure identifies the local external effects of replacing conventional operations.

3.1 Welfare margins of fleet replacement

Consider a city that provides bus service over a fixed route network. Let K denote total bus-kilometers supplied per period and $s \in [0, 1]$ the share of those kilometers operated with electric buses; electrification is an increase in s . The technology mix determines a vector of service attributes $q(s)$ (vehicle age and comfort, noise inside and outside the vehicle, passenger capacity per kilometer, reliability) and, given the regulated fare p , ridership $R(p, q)$. The mix also determines the quantities of externalities generated by bus operations, $X_j(s, K)$, where j indexes local air pollutants, ambient noise, and carbon. Writing $C(s, K)$ for the resource cost of provision and φ_j for the marginal external damage of X_j in money terms, social welfare per period is:

$$W(s) = \underbrace{CS(p, q(s))}_{\text{rider surplus}} + \underbrace{pR(p, q(s)) - C(s, K)}_{\text{fare revenue net of resource costs}} - \underbrace{\sum_j \varphi_j X_j(s, K)}_{\text{external costs}}, \quad (1)$$

where CS is consumer surplus and transfers between the government and operators net out. Differentiating with respect to s , holding K fixed for clarity, the welfare effect of electrification is

$$\frac{dW}{ds} = \underbrace{\frac{\partial CS}{\partial q} \frac{dq}{ds} + p \frac{dR}{ds}}_{\text{transit-market effects}} - \underbrace{\frac{\partial C}{\partial s}}_{\text{resource costs}} - \underbrace{\sum_j \varphi_j \frac{dX_j}{ds}}_{\text{external effects}}. \quad (2)$$

The three terms collect the transit market, the resource cost of the technology switch (the procurement premium for vehicles and charging infrastructure, net of energy and maintenance savings), and the externalities of bus operations. In practice the policy also changes K and other service attributes because new contracts alter fleet composition and operators adjust deployment. We therefore treat service supply and passenger demand as outcomes rather than holding them mechanically fixed, while the externality designs condition explicitly on

total bus activity to isolate technology replacement.

Equation (2) organizes the paper. Section 4 estimates the transit-market response to the new contracts and documents how realized electric deployment covaries with vehicles, kilometers, capacity, speed, and ridership. Section 5 estimates dX_j/ds for the three local externalities: air pollution (Section 5.1), and ambient noise (Section 5.2). Section 6 adds no separate welfare term: it asks whether the pollution reductions reappear downstream, in respiratory emergency visits and school absenteeism, at magnitudes consistent with published concentration–response relationships (Dardati et al., 2024; Chung et al., 2025), disciplining the pollution valuation rather than introducing a new margin.

Two components of Eq. (2) come from institutions and prices rather than from quasi-experimental variation. The first is the resource-cost term, which in Santiago’s implementation is approximately zero because the re-tenders offset the capital premium of the electric fleet with lower energy and maintenance costs (Section 2.3). If the offset is exact, the sign of the welfare effect is determined entirely by the responses we estimate. The second is the carbon term, which is set by the emissions intensity of the national grid. Combining our estimates with damage valuations and grid data into a full welfare evaluation, in the spirit of Hahn et al. (2026), requires no further behavioral estimates; Section 7 performs this aggregation.

3.2 Two estimands

Because electrification is implemented through contract renewal and because operators retain discretion over realized deployment, the paper distinguishes two causal estimands. The first is a *package effect*: the change in outcomes generated by entry into a newly tendered service contract. The package combines electric vehicles with fleet renewal, different vehicle capacities and amenities, charging and maintenance arrangements, and changes in operator deployment. No government replaces drivetrains while holding every other attribute of service fixed. The contract-level effect is therefore the relevant policy object for the transit-market response in Section 4. We also use the same contractual chronology as a complementary reduced-form design for the spatial outcomes below, mapping each contract’s fixed route package into predetermined local exposure around monitoring stations and health centers. These estimates provide an independent check based on contract timing, but because the new contracts bundle electrification with fleet renewal and other operational changes, they identify the package effect rather than the propulsion-technology effect in isolation.

We additionally report how route outcomes covary with the realized electric share of

bus-kilometers. These intensity regressions are informative about how operators deploy the new technology in equilibrium, but they are not a separate causal estimand. Conditional on contract timing, operators still decide where and when to place electric buses, and those choices may respond to demand or other contemporaneous service conditions. We therefore use realized-share regressions to characterize deployment patterns and interpret the contract-start estimates as the cleaner evidence on the causal effect of the implemented package.

The second causal estimand is a *replacement effect*: the change in outcomes at a fixed location when electric bus-kilometers replace conventional bus-kilometers nearby, holding total nearby bus activity constant. Externalities accrue to places, so the natural unit of analysis is a location exposed to the operations of many routes. Conditioning on total activity separates technology composition from service scale: because bus-kilometers themselves respond to electrification, a location-level regression without this control would confound cleaner buses with more or fewer buses. The replacement effect is the estimand for pollution, noise, and the downstream outcomes, and it is the object that corresponds to dX_j/ds in Eq. (2).

3.3 A common design

The empirical analysis combines two sources of variation: the predetermined timing of the new operating contracts and realized changes in electric-bus exposure. The former identifies the effect of the contract package; the latter identifies how outcomes change as route-level operations become more electric. We use the contractual chronology directly for the transit-market analysis and, as a complementary reduced-form design, translate the same contract shocks into predetermined spatial exposure for the externalities and downstream outcomes.

For the transit market, let c index a newly tendered service unit, r a route lineage assigned to that unit, and τ event time relative to the formal start of the new contract. We compare the affected route package with route lineages that remain outside the newly tendered units using a stacked event-study design with route-lineage-by-stack and calendar-week-by-stack fixed effects. Because most contracts begin on Saturdays, we omit the partial calendar week containing the formal start and define event week 0 as the first complete post-contract week. A parsimonious version allows a discrete change at contract start followed by a linear phase-in and reports the implied effect in complete event week 9. This design identifies the reduced-form effect of entering the new contract package on electrification, service supply, operating performance, and passenger demand.

For outcomes observed at fixed locations, we exploit the fact that each new contract affects some locations more than others. When contract c begins, locations near many of its routes receive a large change in exposure, while locations near few of its routes receive a small one. We measure this predetermined exposure by Z_{uc} , the share of bus-kilometers near location u that is operated by contract c 's route package during complete event weeks -9 through -1 . Because the measure is constructed entirely before the contract starts, subsequent changes in service do not affect it. We express Z_{uc} in 10 percentage-point units.

We then compare changes in outcomes around each contract start across locations with different values of Z_{uc} . Each contract defines a separate stack, the partial calendar week containing the contract start is omitted, and we estimate:

$$Y_{uct} = \theta_1 Z_{uc} \mathbf{1}\{\tau_{ct} \geq 0\} + \theta_2 Z_{uc} \tau_{ct} \mathbf{1}\{\tau_{ct} \geq 0\} + \alpha_{uc} + \delta_{ct} + \varepsilon_{uct}, \quad (3)$$

where α_{uc} are location-by-contract fixed effects and δ_{ct} are contract-stack-by-time fixed effects. The first interaction captures the change associated with the contract start, while the second allows the effect to build linearly over subsequent complete weeks. Outcome-specific applications add further fixed effects or controls where appropriate.

Our main reduced-form estimate is $\theta_1 + 9\theta_2$, the implied effect in complete event week 9 for a 10 percentage-point increase in predetermined exposure to the contract package. We estimate the same specification with the local electric share as the dependent variable to document the corresponding first stage. We do not control for contemporaneous total bus activity in these regressions because the contract may itself change the amount of service supplied, making total activity a post-treatment outcome. Finally, because identification comes from only 11 contract starts or fewer, depending on outcome coverage, we complement conventional clustered inference with jackknife inference and report pre-contract trend and late-preperiod placebo tests.

The realized-exposure specifications answer a different question. For route outcomes, they take the form:

$$Y_{ut} = \beta \textit{Electric}_{ut} + \alpha_u + \delta_t + \varepsilon_{ut}, \quad (4)$$

where $\textit{Electric}_{ut}$ is the realized electric share of bus-kilometers on a route. These regressions exploit within-route changes in electric deployment relative to other routes on the same date. Because operators retain discretion over realized deployment, we use them to characterize

equilibrium deployment relationships rather than as the primary causal estimate of the contract package.

For externality and downstream outcomes, we instead estimate the replacement specification

$$Y_{ut} = \beta \textit{Electric}_{ut} + \gamma \textit{Total}_{ut} + \alpha_u + \delta_t + \varepsilon_{ut}, \quad (5)$$

where u is a fixed location and $\textit{Electric}_{ut}$ and $\textit{Total}_{ut}$ measure electric and total nearby bus-kilometers. Here β is the replacement effect: the change associated with substituting electric for conventional bus activity while holding total exposure constant. This is the technology-specific estimand used to quantify pollution and noise effects and, ultimately, to construct the welfare calculation. Each application adapts Eq. (5) to the frequency and structure of the outcome, including richer fixed effects, weather controls, PPML for counts, or event-window specifications.

Treatment exposure is built from the same administrative source throughout: the universe of bus trips, matched to vehicle technology through license plates and to space through successive official route geometries. Figure 3 maps the bus route network and the locations of the health centers, schools, air-quality monitors, and noise monitors used in the analysis. For fixed locations, we draw buffers around each unit and allocate route-level bus-kilometers in proportion to the route geometry lying inside the buffer. The same route packages are used to construct Z_{uc} in Eq. (3), but the exposure weights are fixed using only pre-contract operations. Buffer radii follow the spatial reach of each outcome and are stated in the corresponding sections.

The identifying assumptions differ across the two designs. The contract reduced form requires that locations more exposed to a particular fixed route package would not have experienced differential changes around that contract’s predetermined start date absent the re-tender. Contract expiration dates were inherited from arrangements fixed before electrification entered the policy agenda, and the applications probe this assumption using pre-contract trends, placebo timing, and leave-one-contract-out estimates. The replacement design requires that, conditional on location and time fixed effects and total nearby bus activity, changes in electric exposure are unrelated to location-specific shocks to the outcome. The contractual sequencing is an important underlying source of those changes, but realized deployment also contains operator discretion, which is why we report the contract reduced forms as a separate identification check.

The two designs therefore provide complementary evidence rather than interchangeable estimates. The replacement specification isolates propulsion technology while holding service scale fixed and is the appropriate object for translating estimated externality changes into welfare. The contract reduced form uses a more predetermined source of variation but captures the entire re-tendered package, including fleet renewal and any induced changes in service. Agreement between the two strengthens the interpretation of the replacement estimates; disagreement can reveal either endogenous realized deployment or effects of other components of the contract package.

4 Contract-Driven Electrification and Service Response

This section studies the transit-market response to electrification, the first term of Eq. (2). Figure 2 documents the deployment margin at the system level: by June 2026, electric buses accounted for 55 percent of the operating fleet and roughly two-thirds of daily bus trips. We first establish the link between the contractual chronology described in Section 2.3 and realized electric-bus operation. We then examine how service supply and passenger use evolve as routes become more electric. Because the realized electric share is itself partly chosen by operators, we complement these route-level intensity regressions with reduced-form estimates based directly on the start of the new contracts. The two specifications distinguish changes that accompany realized electrification from those that can be attributed more directly to the contract package that induced it.

4.1 Data construction

We construct a daily panel that follows bus routes between January 2022 and June 2026. The starting point is administrative information provided by the Ministry of Transportation for the universe of bus trips in Santiago. For each trip, the data include the route identifier, bus license plate, date of operation, start and end times, and distance traveled, from which we infer average commercial speed. Route identifiers distinguish main routes and their variants, allowing us to retain the full operational structure of the network. Our final route-day panel covers approximately 1,300 operational route identifiers.⁶

⁶Peak and off-peak periods follow the official time categories of the Ministry of Transportation. Peak hours are 6:30–8:30 a.m. and 6:00–8:00 p.m. on workdays; off-peak hours are 9:30 a.m.–12:00 p.m., 2:30–5:00 p.m., and 9:30–10:30 p.m. on workdays. Weekend observations include all hours on Saturdays, Sundays, and holidays.

We combine the operating records with two additional administrative datasets. First, trip-level ridership data, also provided by the Ministry of Transportation, record the number of passengers transported by each bus trip. Second, we merge trips with the administrative fleet registry using the license plate observed in the operating records. The registry identifies the technology and passenger capacity of each bus, allowing us to distinguish electric from conventional vehicles. The fleet contains approximately 7,500 buses, but not every vehicle operates each day, operators retain discretion over where and when to deploy them, and a bus may serve several routes within the same day. We therefore measure electrification using realized operations rather than fleet availability. For each route-day-time cell, the electric share is electric bus-kilometers divided by total bus-kilometers operated in that cell.

We aggregate trips to the route-day-time level, separately for workday peak hours, workday off-peak hours, and weekends and holidays. For each cell, we compute the number of distinct buses deployed, total bus-kilometers, passenger-capacity-kilometers, average operating speed, usage, and ridership. Bus-kilometers are the total kilometers traveled by all buses serving the route in the corresponding period. Passenger-capacity-kilometers equal the sum across trips of vehicle capacity, including seated and standing passengers, multiplied by kilometers traveled and divided by one million. Average speed is based on total distance traveled relative to operating time. Usage is validations divided by available passenger capacity, expressed as a percentage, and ridership is the number of paying passengers transported in the cell. These definitions preserve the underlying trip-level quantities before aggregation and ensure that buses operating multiple trips or multiple routes are counted consistently.

The contract analysis additionally requires following the same bundle of services when route identifiers change around a re-tender. We therefore link operational identifiers across successive versions of the official route network using administrative crosswalks and route geometries, constructing stable directional route lineages. The route-network vintages cover the full analysis period through June 2026. We then assign each lineage to the service unit operating it after the corresponding contract begins. This construction produces a weekly panel in which the routes transferred under each of the 11 new contracts can be followed consistently before and after formal operation begins. The route-day panel is used for the realized-electric-share specifications, while the route-lineage-week panel is used both to establish the contractual first stage and to estimate reduced-form effects of the contracts on service and passenger outcomes.

4.2 Contract starts and electric-bus deployment

The contractual chronology supplies the institutional source of variation we use. As described in Section 2.3, the expiration dates inherited from the 2007 concessions and the contractual arrangements adopted in 2011 were determined before fleet electrification entered the policy agenda. Later tenders required electric vehicles, so the formal start of a newly tendered service unit should generate a change in realized electric operation. Figure 4 examines this first stage around the 11 contract starts observed during our sample. Because most contracts begin on a Saturday, the calendar week containing the formal start combines operations under the old and new arrangements and is omitted. Event week -1 is therefore the final complete pre-contract week, and event week 0 is the first complete post-contract week.

Panel A of Figure 4 presents an equal-weighted average across tendered service units. For each unit, we compute the electric share of bus-kilometers operated by its future route package and normalize it to the value in event week -1 . Panel B reports estimates from a stacked event-study design. Each contract defines a separate stack containing the route lineages transferred to that service unit and route lineages that are never assigned to a newly tendered unit during the sample. The specification includes stack-by-route-lineage and stack-by-calendar-week fixed effects, is weighted by pre-contract route-lineage bus-kilometers, and uses standard errors two-way clustered by contract and route lineage. Thus, the estimates compare changes in the affected route package with contemporaneous changes among routes that remain outside the new contracts.

The two panels deliver closely aligned patterns. Electric deployment begins to rise during the contractual transition period and accelerates sharply once the new service unit formally starts operating. In the raw data, the electric share increases immediately after the contract begins and continues rising during the following two months. The stacked estimates show the same pattern: the first complete post-contract week produces a discrete increase, followed by a sustained phase-in through event week 9. The correspondence between the unadjusted and regression-adjusted paths indicates that the result is not driven by differential route size, common calendar shocks, or the particular weighting of the stacked estimator. At the same time, movement during the final pre-contract weeks makes clear that implementation is a transition rather than an instantaneous replacement at the formal start date.

Table 1 summarizes this dynamic response using a parsimonious specification with a discrete contract-start jump and a linear weekly phase-in thereafter. The electric share increases by 12.17 percentage points in the first complete post-contract week and by an

additional 3.02 percentage points per week. The implied effect in event week 9 is 39.35 percentage points, relative to a pre-contract mean electric share of 16.07 percent. The remaining columns decompose the operational change. By event week 9, electric bus-kilometers have risen by 42.19 percent of pre-contract total bus-kilometers, while non-electric bus-kilometers have fallen by 36.24 percent. Total bus-kilometers increase by 5.95 percent. Electric deployment therefore consists primarily of conventional service being replaced by electric service, although the new contracts also generate a modest expansion in total operations. This decomposition directly documents the distinction developed in Section 3.2: the policy is predominantly a technology replacement, but the package implemented through the re-tenders also changes service scale.

4.3 Service supply and passenger response

We use two complementary specifications to study service supply and passenger response. The first follows realized electrification within routes. Let r denote a route, d a date, and h one of three time samples: workday peak hours, workday off-peak hours, or weekends and holidays. We estimate the following panel data regression model:

$$Y_{rdh} = \beta S_{rdh} + \phi_r + \theta_d + \varepsilon_{rdh}, \quad (6)$$

where S_{rdh} is the share of bus-kilometers operated by electric buses, ϕ_r are route fixed effects, and θ_d are date fixed effects. A unit change in the treatment corresponds to moving from no electric operation to fully electric operation in the cell. Because operators retain discretion over where and when electric buses are deployed, realized electric shares may respond to contemporaneous demand or other operating conditions. We therefore interpret β as an equilibrium deployment relationship rather than as the causal effect of electrification itself.

The second specification uses contract starts directly. We apply the stacked design described in Section 3.3 to each service and passenger outcome, comparing affected route lineages with never-treated lineages over complete event weeks -9 through 9 . As in the first-stage analysis, the partial calendar week containing the contract start is omitted. Table 2 summarizes the short-run response using a contract-start jump and linear post-contract phase-in and reports the implied effect in event week 9. Figures A.1, A.2, and A.3 report the corresponding unrestricted event-study paths. Some outcomes begin adjusting before the formal contract date, consistent with implementation being a contractual transition rather than an instantaneous treatment, so we interpret the event-date coefficients with caution.

For the realized-share specification, all outcomes except bus-kilometers are weighted by average route bus-kilometers during the first 60 observed days, and standard errors are clustered by route. In the contract specification, all outcomes except bus-kilometers are weighted by pre-contract bus-kilometers, and standard errors are two-way clustered by contract and route lineage. Ridership enters in logarithms and other outcomes in levels.

Table 2 first shows that realized electric deployment is strongly associated with greater service supply. Moving from no electric operation to fully electric operation is associated with 2.23 additional buses during peak hours, 4.44 during off-peak hours, and 4.20 during weekends and holidays. Bus-kilometers increase by 19.34, 22.10, and 31.51 kilometers, respectively. Passenger-capacity-kilometers instead fall by 0.013, 0.017, and 0.039 million, while average operating speed is essentially unchanged. Routes and periods with more intensive electric deployment therefore use more vehicles and operate more kilometers, but do not provide greater measured passenger-carrying capacity or faster service.

The contract estimates show a related but more selective service response. By event week 9, bus-kilometers increase by 28.13 kilometers during peak hours, 36.72 during off-peak hours, and 91.24 during weekends and holidays. The number of buses rises significantly only on weekends and holidays, by 3.23 vehicles, while capacity-kilometers show no statistically detectable change. Commercial speed falls modestly, by roughly 0.5 km/h in each period. The unrestricted event studies show that some of these service adjustments begin before the formal contract start, but the broad pattern around the transition is an increase in bus-kilometers rather than in passenger capacity.

The sharpest difference between the two specifications concerns passenger demand. Realized electric shares are positively associated with usage—by 11.68, 10.12, and 9.96 percentage points across peak, off-peak, and weekend periods—and with ridership, by approximately 4.5, 6.4, and 4.7 percent. The contract estimates do not reproduce this pattern. By event week 9, usage changes by -1.99 , -0.68 , and -0.79 percentage points, while ridership changes by approximately -4.0 percent, zero, and 0.7 percent; none of these estimates is statistically different from zero. The event-study paths likewise show no sustained positive passenger response following the contractual transition. Thus, while routes with more intensive electric deployment carry more passengers, the contract-induced shift toward electric operation does not generate a detectable short-run increase in passenger demand.

4.4 Discussion and interpretation

The combined evidence separates two objects that would otherwise be easy to conflate. The contractual transition clearly induces electrification: Figure 4 and Table 1 show large substitution from conventional to electric bus-kilometers and a modest increase in total operations. Table 2 further shows that the new contracts expand service, particularly bus-kilometers, without generating a detectable short-run increase in passenger demand. The appendix event studies qualify a sharp before-and-after interpretation because some service outcomes begin adjusting before the formal start date, but this is consistent with the institutional process: buses, depots, operators, and route packages transition gradually into the new contract. The contract design therefore captures the short-run response to an implementation process rather than to a single instantaneous technological switch.

The realized-share regressions reveal a different pattern. Routes and periods in which operators deploy electric buses more intensively have more buses and bus-kilometers and also exhibit higher usage and ridership. Yet the contract transition itself produces no comparable passenger increase. This divergence suggests that the positive passenger coefficients reflect endogenous deployment or other adjustments correlated with electric operation. Operators may assign electric buses more intensively to routes or periods where demand is stronger or improving, or may change schedules and other operating margins jointly with electric deployment. The data do not distinguish among these channels, but they do rule against interpreting the realized-share coefficient as a clean demand effect of electric buses.

This distinction matters for the policy estimand. The contracts combine fleet electrification with vehicle renewal, charging and maintenance arrangements, operator changes, and changes in service provision. The relevant policy object is therefore the implemented contract package rather than the drivetrain in isolation. At the same time, the absence of a contract-induced ridership response means that we should not attribute the positive realized-share relationship to quieter, newer, or more comfortable buses themselves. Unlike bus-priority infrastructure, where passenger gains operate through faster travel times (González and Silva, 2025), the Santiago re-tenders substantially change fleet technology and service provision without producing a detectable short-run increase in ridership. For the welfare analysis in Section 7, we therefore distinguish the contract-based transit-market response from the broader equilibrium associations observed in realized electric deployment.

5 Externalities

We now turn to the external effects of electrification, the third term of Eq. (2). This section estimates replacement effects for two local externalities of bus operations: air pollution and ambient noise. These are changes in outcomes at fixed locations when electric bus-kilometers replace conventional ones, holding total bus activity fixed.

5.1 Air pollution

The air-quality analysis uses the official monitoring network, which records several pollutants hourly at a small number of fixed locations. This setting is particularly useful for studying a local effect of bus electrification because we observe both the intensity of bus operations around each monitor and the hourly wind conditions that determine how informative the spatial position of nearby routes is for measured pollution. We therefore combine two sources of variation. The first is the composition of nearby bus traffic: conditional on total bus-kilometers within a fixed radius, an increase in electric bus-kilometers corresponds to replacing conventional bus activity with electric activity. The second is whether those nearby routes are upwind or downwind of the monitor in a given hour. The resulting design remains centered on the replacement effect defined in Section 3.2, while using the wind information to distinguish the local composition effect from an additional directional component. Figure A.4 illustrates the construction using actual route geometries around two monitoring stations. For a given wind realization, routes located on the upwind side of a monitor receive positive signed exposure, because emissions are carried toward the station, whereas routes on the opposite side receive negative exposure. The three circles show the 500, 750, and 1,000 meter radii used to characterize nearby bus activity.⁷

Data and specification. We combine validated hourly pollution readings from January 2022 through December 2025 with hourly route-level operations and hourly wind speed and direction. We study $\text{PM}_{2.5}$, PM_{10} , CO , NO_2 , and NO_x . The sample contains seven monitoring stations, with the number of usable station-hours varying across pollutants because of pollution-data availability. Our baseline measure of nearby activity uses a 750-meter radius. For each monitor, route geometry, and hour, let Δ_{irh} denote the angular

⁷Satellite pollution data are monthly and spatially smoothed and therefore cannot be aligned with the hourly operations and wind variation used here. Combining the two approaches would consequently mix a sharply local short-run design with a substantially coarser spatial and temporal estimand.

difference between the bearing from monitor i toward route segment r and the meteorological direction from which the wind is blowing. We define signed wind exposure by multiplying route-level bus-kilometers by wind speed and $\cos(\Delta_{irh})$. A positive value therefore corresponds to bus activity upwind of the monitor, a negative value to activity downwind, and the magnitude tends continuously to zero as wind speed falls or the route becomes cross-wind. Importantly, calm and low-wind observations remain in the sample, the signed exposure measure places little directional weight on such hours. We construct this measure separately for electric and total bus activity. The main estimating equation for pollutant p is:

$$y_{ipth} = \beta_p E_{ith} + \gamma_p B_{ith} + \theta_p S_{ith}^E + \delta_p S_{ith}^B + \alpha_{itp} + \eta_{iwp} + \kappa_{isp} + \mu_{iqp} + \varepsilon_{ipth}, \quad (7)$$

where E_{ith} and B_{ith} are electric and total bus-kilometers within 750 meters of monitor i , and S_{ith}^E and S_{ith}^B are the corresponding signed wind exposures. The coefficient β_p is therefore the local replacement effect of shifting nearby bus activity from conventional to electric, holding total nearby bus activity fixed, while θ_p captures whether that replacement effect changes when electric bus activity is more strongly upwind rather than downwind. The fixed effects α_{itp} absorb station-by-date conditions, η_{iwp} absorb station-specific hour-of-week patterns, κ_{isp} absorb station-by-eight-sector wind-direction differences, and μ_{iqp} absorb station-by-wind-speed-bin differences. Wind-speed bins are defined separately by station from the distribution of observed wind speeds and are constructed without reference to pollution outcomes. Bus activity enters in units of 10 kilometers and signed wind exposure in units of 10 kilometer-meters per second. Standard errors are clustered by monitoring station and to adjust for the small number of clusters we report two-sided restricted wild cluster bootstrap p -values from 9,999 replications using Webb weights (Cameron et al., 2008; Webb, 2023).

Results. Table 3 reports the baseline hourly estimates. The local replacement coefficient is negative for PM_{2.5}, PM₁₀, and NO₂, whereas the estimates for NO_x and CO are close to zero. An additional 10 electric bus-kilometers, holding total nearby bus activity fixed, reduces PM_{2.5} by 0.064 $\mu\text{g}/\text{m}^3$, PM₁₀ by 0.322 $\mu\text{g}/\text{m}^3$, and NO₂ by 0.109 ppb. The corresponding clustered p -values are 0.088, 0.037, and 0.086, while the Webb bootstrap p -values are 0.094, 0.047, and 0.079. Thus the substantive conclusions are similar under the two inference procedures: the PM₁₀ effect is estimated relatively precisely, while the PM_{2.5} and NO₂ estimates provide somewhat weaker evidence in the same direction. By contrast, the NO_x coefficient is 0.027 and the CO coefficient is essentially zero, with neither estimate different from zero. The signed electric-wind coefficients are also small and statistically close to

zero for all five pollutants. The data therefore support a local fleet-composition effect but provide little evidence that this effect becomes detectably stronger when the relevant routes are more sharply upwind of the monitor. This pattern is plausible in a setting with generally modest wind speeds: nearby emissions can affect local concentrations through short-distance dispersion and urban mixing even when wind direction does not generate a sufficiently sharp hourly plume contrast to identify a separate directional effect.

The magnitudes are economically meaningful but not implausibly large. A mechanical linear extrapolation that replaces the average remaining conventional bus activity within 750 meters with electric activity implies reductions of approximately $0.7 \mu\text{g}/\text{m}^3$ in $\text{PM}_{2.5}$, $3.6 \mu\text{g}/\text{m}^3$ in PM_{10} , and 1.2 ppb in NO_2 . Relative to sample means, these correspond to declines of roughly 3, 6, and 6 percent, respectively. These magnitudes are of the same broad order as the emerging evidence from passenger-vehicle electrification. For example, Baran et al. (2026) estimate that a one-standard-deviation increase in electric-vehicle penetration reduces average NO_2 concentrations by approximately 3 percent in their preferred specification. The comparison is necessarily approximate because our treatment replaces intensive bus activity in the immediate vicinity of a monitor, whereas their variation comes from county-level passenger-vehicle adoption. The somewhat larger local NO_2 effect implied by complete bus electrification is therefore not surprising. The smaller proportional $\text{PM}_{2.5}$ response is also consistent with the fact that electrification eliminates tailpipe emissions but leaves tire wear, road abrasion, resuspended dust, and other non-exhaust particulate sources largely intact. PM_{10} displays the largest estimated proportional reduction in our data, and we consequently interpret its magnitude as the overall effect of replacing the conventional bus fleet rather than as evidence that all of the response operates through tailpipe particulate emissions alone.

Appendix Table A.1 reports two complementary sets of checks. First, we re-estimate the local replacement specification without requiring wind observations. This expands the 2022–2025 samples by roughly 8–9 percent and omits the signed-wind terms and wind fixed effects. The PM_{10} coefficient remains negative, while the $\text{PM}_{2.5}$ and NO_2 coefficients attenuate substantially and the CO estimate becomes positive. The additional observations therefore increase sample size but do not simply increase precision, suggesting that hours with observed wind constitute a somewhat different and more informative comparison rather than merely a smaller version of the full sample. Second, we extend both specifications through the available 2026 observations. Because validation of the 2026 pollution data is incomplete, these extensions use validated readings when available, preliminary readings otherwise, and not-yet-validated readings when neither of the former is available. In the wind specification,

the NO_2 coefficient remains negative and similar in magnitude, the PM_{10} coefficient remains negative but becomes less precise, and the $\text{PM}_{2.5}$ coefficient moves toward zero. We therefore treat these estimates as provisional robustness evidence rather than as part of the baseline sample. Across all of these specifications, the signed electric-wind coefficient remains close to zero. Taken together, the results indicate that replacing conventional buses with electric buses improves local air quality, with the clearest evidence for particulate matter and NO_2 .

As a complementary identification check, Table A.2 replaces realized electrification with the predetermined start dates of the tendered service contracts. For each monitoring station and contract, we measure exposure using the share of nearby bus-kilometers supplied by the contract’s fixed route package during the nine complete pre-contract weeks, so that neither treatment timing nor exposure uses post-contract operations. Contract starts generate a clear first stage: a 10 percentage-point increase in predetermined contract exposure raises the local electric share by about 5.3 percentage points by event week 9. Over the same horizon, $\text{PM}_{2.5}$, PM_{10} , NO_x , and CO decline, while the NO_2 estimate is close to zero; the negative estimates are also stable to sequentially omitting individual contracts, and pre-contract trend and placebo tests provide little indication of differential changes before treatment. Because the new contracts bundle electrification with broader fleet renewal and operational changes, these reduced-form estimates do not identify the pure electric-bus replacement effect. Instead, they provide independent evidence that the institutional reforms generating the transition to electric buses were followed by improvements in local air quality.

5.2 Noise

Data and specification. We construct an hourly panel of ambient noise from the Ministry of the Environment’s monitoring network, obtained through a FOIA request: equivalent sound pressure levels in dB(A) for 12 stations, January 2022 through December 2025. We match each station-hour to electric and total bus-kilometers operating within 50, 100, and 200 meter buffers, using the same hourly bus operations and successive official route geometries as in the pollution analysis. The narrow buffers reflect the local nature of traffic noise and allow the effect to vary with distance from bus corridors.

We apply the two designs described in Section 3.3. The replacement specification relates hourly noise to electric and total nearby bus-kilometers, measured in units of 10 km, with station-by-date and station-by-hour-of-week fixed effects. The electric-bus coefficient therefore measures the change in dB(A) associated with replacing 10 nearby conventional bus-kilometers

in a given hour with electric bus-kilometers while holding total activity fixed. We cluster standard errors by station and, with only 12 monitors, report restricted wild cluster bootstrap p -values using Webb weights and 9,999 replications. As a robustness check, we additionally absorb exact date-by-hour fixed effects. The complementary contract reduced form maps each re-tendered contract’s fixed route package into predetermined station exposure using complete event weeks -9 through -1 , estimates Eq. (3) at the hourly frequency, and uses delete-one-contract jackknife inference.

Results. Panel A of Table 4 reports the replacement estimates. The effect of replacing 10 conventional bus-kilometers with electric operation is -3.74 dB(A) within 50 meters, -1.32 dB(A) within 100 meters, and -0.60 dB(A) within 200 meters. The corresponding wild cluster bootstrap p -values are 0.072, 0.106, and 0.093. Because average bus activity rises mechanically with buffer size, evaluating these coefficients at mean nearby activity implies full-electrification effects of approximately -0.80 , -0.89 , and -0.92 dB(A), respectively. The stronger time controls in Appendix Table A.3 retain negative coefficients at all three distances, with implied effects around one-half dB(A).

Panel B reports the contract reduced form. A 10 percentage-point increase in predetermined contract exposure raises the local electric share by 2.6 percentage points within 50 meters and lowers ambient noise by 0.137 dB(A) by event week 9; the contract-jackknife p -value is below 0.001. At 100 meters the corresponding reduced-form effect is -0.131 dB(A) ($p = 0.004$), while at 200 meters it is essentially zero. The 50- and 100-meter estimates exhibit neither differential pre-trends nor late-preperiod placebo effects and remain negative when each identifying contract is omitted in turn. The two designs therefore point to the same conclusion: electrification reduces ambient noise in the immediate vicinity of bus routes, with the marginal effect of nearby electric bus activity attenuating sharply with distance.

Interpretation. Vehicle-level studies find larger differences, as expected. Electric buses are about 5 dB(A) quieter than diesel buses during acceleration (Borén, 2020), while replacing diesel-hybrid buses with electric buses cuts indoor low-frequency sound during passages by up to 10 dB (Glebe et al., 2024). Our outcome is outdoor A-weighted ambient noise, which combines bus passages with all other sound sources. An ambient reduction on the order of one dB(A) is therefore a plausible diluted counterpart of these source-level differences and may understate perceived gains for residents immediately along bus corridors.

6 Downstream Effects

The externality estimates establish that electrification reduced local air pollution along the network. This section asks whether those reductions reappear downstream, in respiratory emergency visits and school absenteeism. These outcomes add no separate welfare term: they are implications of the pollution term under a concentration–response mapping, and we use them to discipline magnitudes. The published estimates that anchor this mapping imply effects that are small relative to the variation in these outcomes, so we present the evidence as a validation exercise. The question is whether the downstream data are consistent with the pollution channel, not whether they can independently detect it.

6.1 Emergency respiratory visits

We first ask whether the pollution reductions are reflected in emergency-room visits for respiratory conditions. The records identify the health center where the patient was treated, not the patient’s residence, so we assign exposure using bus service around the center. This is a natural approximation if patients visit nearby centers, but it can also capture changes in health-center utilization, catchment areas, or access to care. The emergency-room estimates are therefore suggestive evidence on downstream morbidity, and no more than that.

We construct a balanced weekly panel of the 95 public emergency centers located within 2 km of the route network, from January 2022 through June 2026. For each center and week, we aggregate visits for respiratory causes and construct a non-respiratory comparison outcome as total visits minus respiratory visits. Our preferred specification defines high electric-bus adoption as the first week in which the electric share of bus-kilometers within 2 km of the center reaches 20 percent.⁸ We then estimate event-window regressions:

$$Y_{it} = \beta_1 \mathbf{1} 0 \leq \tau_{it} \leq 52 + \beta_2 \mathbf{1} \tau_{it} > 52 + \gamma TotalKM_{it} + \alpha_i + \lambda_t + \varepsilon_{it}, \quad (8)$$

where τ_{it} is the number of weeks since health center i first reached high adoption, $TotalKM_{it}$ is total bus-kilometers within 2 km of the center, and α_i and λ_t are health-center and week fixed effects. For count outcomes we estimate the model by PPML; we also estimate

⁸The event-window estimates require roughly one year (49 weeks) of pre-treatment observations, so we drop 26 treated centers that crossed the 20 percent threshold too early in the panel. The estimating sample contains 69 centers, of which 68 reach high adoption and one never does. The first crossing among the retained treated centers occurs in the week of January 2, 2023.

specifications for the respiratory share of visits and for the log ratio of respiratory to non-respiratory visits.

Table 5 reports the results. The point estimates for respiratory visits are negative, falling by 0.9 percent during the first year after high adoption and by 5.4 percent thereafter, but are imprecisely estimated. The non-respiratory comparison outcome also declines, by 0.6 percent during the first year and by 3.3 percent thereafter. The respiratory share falls by 0.2 percentage points in the first year and by 0.5 percentage points thereafter, while the log respiratory-to-non-respiratory ratio falls by 2.1 and 4.7 percent, respectively. None of these estimates is statistically distinguishable from zero, and the pattern provides little evidence of a respiratory-specific effect.⁹ The event-study estimates in Appendix Figure A.5 show no strong differential pre-trends, but the post-adoption coefficients remain noisy and reveal no sharp respiratory-specific break.¹⁰

We therefore do not interpret the emergency-room results as clean causal evidence of downstream health effects. The estimates are consistent with the pollution channel, but visits measured at the treating facility, with no link to residential exposure, cannot distinguish that channel from changes in emergency-center utilization. The contract-based reduced form reinforces this caution: although contract timing strongly predicts local electrification, its health effects need not isolate propulsion technology because the contracts also change other dimensions of service provision.

6.2 Student absenteeism

We next examine student absenteeism. Children are especially sensitive to air pollution, and school absences provide a broad measure of morbidity that captures responses not severe enough to generate emergency visits. A growing literature finds that pollution increases absences: Currie et al. (2009) show that carbon monoxide raises school absences in Texas, and Chung et al. (2025) estimate that a 10 $\mu\text{g}/\text{m}^3$ increase in daily $\text{PM}_{2.5}$ increases full-day

⁹Appendix Table A.4 uses the predetermined start dates of the 11 tendered service-unit contracts as an alternative source of variation. Contract exposure strongly predicts subsequent electrification, but the reduced form shows no effect on non-respiratory visits or respiratory composition. Respiratory visits increase by about 1.6 percent; this estimate is robust to leave-one-contract-out inference and shows no detectable pre-trend. Because contract implementation changes several dimensions of service provision besides propulsion technology, we do not interpret this estimate as the causal effect of electrification on respiratory health.

¹⁰Appendix Table A.5 shows that this conclusion is robust to restricting the sample to SAPU centers and to excluding large-volume, low-volume, or unstable centers. Across these samples, respiratory visits tend to decline after high adoption, especially in later weeks, but non-respiratory visits often decline as well.

absences by 5.7 percent. These estimates provide a benchmark for whether the pollution reductions induced by bus electrification are large enough to generate detectable changes in school attendance.

We construct a monthly school-level panel from MINEDUC administrative attendance records, covering regular primary-school students in grades 1–8 from March 2022 through June 2026. The outcome is the school-month absence rate: the number of student-days missed divided by the number of possible student-days, so students are weighted by their possible attendance days. We exclude student-month observations with 100 percent absenteeism before collapsing to the school-month level, since these observations may capture students who transferred or stopped attending a school while remaining in the administrative records. The estimation sample uses schools located within 5 km of at least one bus route, providing a common spatial sample across exposure buffers.¹¹ Exposure is constructed as in the pollution analysis. We allocate bus-level daily kilometers to buffers of 500 meters, 1 km, 2 km, and 5 km around each school according to the fraction of each route inside the buffer, using the complete sequence of route geometries and route-code crosswalks, and aggregate these measures to the school-month level. The specification is the common design of Eq. (4): we regress the absence rate on electric bus-kilometers within each buffer, controlling for total bus-kilometers within the same buffer, with school and month-year fixed effects. Standard errors are clustered by school.

Table 6 reports the results. The coefficients on electric bus-kilometers are negative at all four distances but statistically indistinguishable from zero. Replacing 100,000 monthly bus-kilometers with electric bus-kilometers, holding total bus activity fixed, is associated with changes in the absence rate of -0.664 , -0.300 , -0.051 , and -0.004 percentage points within 500 meters, 1 km, 2 km, and 5 km, respectively. The estimates decline rapidly with distance and are imprecise, and the coefficients on total bus-kilometers are also statistically insignificant. Extending the sample through June 2026 therefore leaves the earlier conclusion unchanged: the realized-electrification design provides no evidence of a detectable response of school absenteeism.

As a complementary design, Appendix Table A.6 exploits the predetermined timing and route packages of the 11 service contracts. For each school and contract, we measure exposure as the share of bus-kilometers within 2 km supplied by the contract’s fixed route package

¹¹School coordinates come from the MINEDUC school directory. Because some schools have missing coordinates in individual years but valid coordinates in others, we append the annual directories and use coordinates from other available years when necessary.

during the nine complete pre-contract weeks. Event time is measured in school months, excluding January, February, July, and December, and the contract-start school month is omitted. Predetermined contract exposure strongly predicts subsequent electrification: a 10 percentage-point increase in exposure raises the electric share of nearby bus-kilometers by 7.0 percentage points by school-event month 4. Yet the corresponding reduced-form effect on absenteeism is only 0.052 percentage points and is statistically insignificant using delete-one-contract inference ($p = 0.296$). The absenteeism pre-trend and late-preperiod placebo tests likewise provide no evidence of differential movements before the contract begins. Thus, an institutional design that generates a strong first stage in local electrification also produces no detectable change in school absences.¹²

These nulls are consistent with the scale of the pollution effects. Using the estimate in Chung et al. (2025), a $1 \mu\text{g}/\text{m}^3$ reduction in $\text{PM}_{2.5}$ implies roughly a 0.57 percent reduction in student absences. At the sample mean absence rate of 13.02 percentage points, this corresponds to only about 0.07 percentage points. Two features of our setting further limit power to detect such an effect. First, our outcomes are monthly and aggregated at the school level, whereas the strongest evidence in the literature comes from daily pollution shocks and daily attendance, so aggregation can dilute short-lived responses. Second, bus emissions are only one component of children’s total exposure: students are exposed to pollution at home, at school, and while commuting, so the change in personal exposure generated by electrifying nearby bus service is smaller than the local pollution reduction measured near bus routes.

Overall, both empirical designs point to the same conclusion. We find no measurable response of primary-school absenteeism to bus electrification, despite a strong contract-induced first stage in local electric-bus exposure. This should not be interpreted as evidence against the pollution channel documented in Section 5.1. Given the magnitude of the estimated $\text{PM}_{2.5}$ reductions and the concentration-response estimates in the absenteeism literature, the downstream effect on monthly school absences is expected to be small and difficult to distinguish from zero.

7 Welfare Synthesis

Section 2.3 established that the re-tenders electrified the fleet at approximately no additional resource cost, so the welfare effect of electrification in Eq. (2) reduces, to first order, to the

¹²Appendix Figure A.6 presents the corresponding dynamic contract-start estimates.

transit-market and external effects estimated in Sections 4 through 6. This section prices those responses. We proceed in two steps. We first state each benefit as the product of one replacement estimate and one price, evaluated at full electrification of the bus activity observed around a monitor. We then scale these products by the realized rollout to obtain the citywide annual flow in each year from 2022 to the first half of 2026.

Three conventions apply throughout. All values are expressed in US dollars of December 2025, converted at the exchange rate of December 31, 2025, and discounted, where a time dimension arises, at the official social discount rate of 5.5 percent.¹³ Wherever an official Chilean social price exists we use it, drawing on the published literature only for dose-response relationships that have no official counterpart. Because the valuation of avoided mortality dominates the total and the two defensible conventions differ by a factor of about four, we report every mortality-driven figure under both: the full statistical-life value (VVE) and a life-years valuation (VLY). For every other modeling choice we state the branch we take and show the alternatives in the sensitivity analysis.

7.1 Prices and parameters

Table 7 collects every price, dose-response coefficient, and quantity used in this section, with sources. The prices are the official social prices of the Chilean public investment-evaluation system (Ministerio de Desarrollo Social y Familia, 2026): a value of a statistical life of 60,000 UF (USD 2.62 million), a social price of carbon of USD 78.8 per ton of CO₂, and social prices of electricity, diesel, and new buses. Using the government’s own appraisal prices anchors the exercise. These are the values against which every Chilean public investment is evaluated. For responses without an official price we use the causal estimates already introduced: the acute mortality response of Deryugina et al. (2019), the emergency-room response of Dardati et al. (2024), and a road-noise hedonic depreciation of 0.40 percent of property value per dB(A), the central value of the meta-analysis literature (Nelson, 2008). The quantities that convert per-location estimates into aggregates come from our own data construction: census population and dwelling counts intersected with the route network, residential values from observed transactions, and the administrative bus-kilometer panel.

¹³USD 1 = CLP 911.18; UF 1 = CLP 39,727.96 = USD 43.60.

7.2 The benefits of replacement

Each benefit is the product of one replacement estimate from Section 5 and one price from Table 7; Panel A of Table 8 collects the resulting citywide annual flows. We state each product once.

Air pollution. The hourly replacement coefficient in Table 3 implies that 10 electric bus-kilometers per hour within 750 meters of a monitor, replacing conventional service, lower $\text{PM}_{2.5}$ by $0.064 \mu\text{g}/\text{m}^3$. Mean bus activity within 750 meters of the seven monitors is 154 bus-kilometers per hour, of which 43 are electric over the sample. Full electrification of this activity therefore lowers $\text{PM}_{2.5}$ by $0.99 \mu\text{g}/\text{m}^3$ relative to an all-conventional fleet. The reduction already realized at the sample mean is $0.27 \mu\text{g}/\text{m}^3$, and electrifying the remaining conventional service would add the $0.7 \mu\text{g}/\text{m}^3$ reported in Section 5.1. We apply this roadside estimate directly to the population living along the network.

The transfer is reasonable. The electric share of bus activity at the monitors over the sample, 28 percent, matches the citywide average of 29 percent over 2022–2025. The 750-meter buffers around routes electrified by 2025 contain 6.1 million residents, most of the population of the Metropolitan Region, so the exposed population is the city at large rather than a few busy corridors, and we take the monitors as representative of it. The estimate is also plausible against source apportionment: motor vehicles contribute 37 percent of ambient $\text{PM}_{2.5}$ in central Santiago (Barraza et al., 2017) and diesel buses about a quarter of transport-sector black carbon (Climate and Clean Air Coalition, 2017), which places the bus contribution near 9 percent, or $1.9 \mu\text{g}/\text{m}^3$ at an ambient level of $20 \mu\text{g}/\text{m}^3$. The full-electrification effect we use is about half of this ceiling.

The realized reduction in each year is the full-electrification effect scaled by the electric share of bus-kilometers: $0.13 \mu\text{g}/\text{m}^3$ in 2022, 0.41 in 2025, and 0.60 at the 2026 running rate. Combined with the acute mortality response of Deryugina et al. (2019), applied to the 1.19 million residents aged 60 and older within 750 meters of electrified routes, these reductions avoid about 39 premature deaths per year in 2022, 123 in 2025, and 178 at the 2026 rate.¹⁴

Appendix A checks this translation against Chilean death records. We compare each bus-served comuna with a synthetic control built from comunas outside the Metropolitan

¹⁴The response is estimated for ages 65 and over; we apply it to the census count of residents aged 60 and older and state the transfer explicitly. Restricting the exposed population to residents within 200 meters of electrified routes, 1.02 million aged 60 and older, lowers the mortality row by 14 percent (Panel B of Table 8).

Region and relate elderly mortality to contract-induced electric exposure. The estimate is imprecisely estimated and statistically indistinguishable from zero. Converted with the contract first stage and applied to the 2025 exposure and baseline, it implies about 250 avoided deaths per year. The 123 deaths implied by the pollution calculation sit well inside the confidence interval. We therefore keep the concentration–response route for valuation. The direct estimate cannot distinguish an effect of this size from zero, but it does not contradict it.

We value these deaths under two conventions and report both throughout. The first applies the full VVE, the statistical-life value used throughout the Chilean investment-evaluation system. Because the VVE is calibrated on prime-age adults while acute pollution mortality falls disproportionately on the frail elderly, this is an upper bound on the mortality benefit. The second follows the life-years metric that Deryugina et al. (2019) emphasize for acute elderly mortality. Their preferred estimates, 2.99 life-years and 0.69 deaths per million elderly per $\mu\text{g}/\text{m}^3$ -day, imply 4.3 life-years lost per death. We price each life-year at USD 163,000, the VVE annuitized at the 5.5 percent social discount rate over a 40-year prime-age horizon, so each avoided death is worth USD 706,000, and the avoided deaths correspond to 170 to 530 life-years per year between 2022 and 2025. The resulting mortality row rises from USD 101 million in 2022 to USD 321 million in 2025 at the VVE, and from USD 27 million to USD 87 million at the VLY; under either convention it is the largest benefit line.¹⁵

Noise. Replacing 10 conventional bus-kilometers per hour with electric service lowers ambient noise by 3.74 dB(A) within 50 meters of a monitor and by 0.60 dB(A) within 200 meters (Table 4). Evaluated at mean bus activity in each buffer, full electrification lowers noise by 0.80 dB(A) at 50 meters and by 0.92 dB(A) at 200 meters. We build the citywide row in two ways. The first applies the 200-meter effect to the residential stock within 200 meters of electrified routes: 1.99 million dwellings with a mean transaction value of about USD 131,000, or USD 259 billion. At a noise depreciation index of 0.40 percent per dB(A), the implied capitalization is $0.40 \times 0.92 = 0.37$ percent of that stock, or USD 53 million per year when annualized at 5.5 percent. The second applies the 50-meter effect to the 800,667 dwellings within 50 meters, a stock of USD 105 billion, and yields USD 18 million per year. Panel A of Table 8 reports both. The totals use the 200-meter figure, which

¹⁵The morbidity row applies the 0.36 percent response of Dardati et al. (2024) to the 734,764 baseline respiratory emergency visits recorded in exposed areas in 2022, priced at the FONASA free-choice tariff for an emergency-medicine consultation, USD 19.9 (FONASA, 2024): between 340 and 1,090 avoided visits per year, under USD 25,000. Because the VVE already prices fatality risk, the morbidity row is additive only under a cost-of-illness reading, which we adopt.

covers the dwellings that face a bus corridor rather than only the first row of buildings; Panel B shows the 2025 total under the 50-meter alternative. Using the median rather than the mean transaction value lowers either stock by one-fifth and the benefit in proportion. Zero remains a defensible floor given the imprecision discussed in Section 5.2. The citywide trajectory scales the full-electrification benefit by the realized electric share.

Carbon. Replacing 10,000 diesel bus-kilometers with electric operation saves a net 15 tons of CO₂: about 18.2 tons of avoided diesel combustion against 3.2 tons from the charging electricity at the current grid emission factor.¹⁶ At the Chilean social price of carbon, USD 78.8 per ton, the central case is about USD 1,180 per 10,000 electric bus-kilometers, in a range of USD 770 to 2,860 as the price moves between USD 51 and USD 190 (Interagency Working Group on Social Cost of Greenhouse Gases, 2021; U.S. Environmental Protection Agency, 2023).

Fare revenue. The contract design detects no ridership response: nine weeks after a contract starts, passenger usage and ridership on the re-tendered routes are not detectably higher (Section 4). We therefore set the fare-revenue row to zero. Routes with a higher realized electric share do carry more riders, but Section 4.4 attributes this association to where operators deploy electric buses rather than to a demand response, so we do not value it. Riders may still gain from quieter and smoother vehicles. Our design does not measure this margin, and we add no consumer-surplus term.

Table 8 converts the per-unit flows into the citywide annual effect of the realized rollout, scaling each replacement effect by that year’s electric share of bus-kilometers or by electric bus-kilometers, and reports the trajectory from 2022 to the first half of 2026 as the electric share of bus service rises from 13 to 60 percent. The 2025 flow is USD 362 million at the VVE and USD 128 million at the VLY; per 10,000 electric bus-kilometers, these flows correspond to about USD 22,400 and USD 7,900. At the 2026 running rate the flows reach USD 524 million and USD 184 million. Mortality accounts for 89 percent of the 2025 total at the VVE and 68 percent at the VLY. The annual state subsidy that closes the operating deficit of the whole system, buses, Metro, and commuter rail combined, was CLP 1,156 billion in 2025, about USD 1.27 billion (Directorio de Transporte Público Metropolitano, 2026). The realized gains from the rollout in 2025 therefore equal close to three-tenths of the annual subsidy at the VVE and one-tenth at the VLY.

¹⁶Diesel: $10,000/1.47 \approx 6,800$ liters at the homologated fuel economy of 1.47 km per liter (International Council on Clean Transportation, 2022), at 2.68 kg of CO₂ per liter. Electricity: $10,000 \times 1.34 = 13,400$ kWh at 0.2384 t CO₂e/MWh, the 2023 year-end factor of the Coordinador Eléctrico Nacional.

7.3 The cost side

The structure of the tenders, described in Section 2.3, implies that these benefits were obtained at approximately no additional resource cost, and the official prices themselves make the point compactly. The capital premium of an electric over a diesel urban bus is USD 118,000 at social prices (247,000 against 129,000). Its energy cost per kilometer is USD 0.176 against USD 0.601 for diesel, a 71 percent energy-cost saving, and the transport authority reports operating-cost savings of 66 percent and maintenance savings of 44 percent for electric operation (Directorio de Transporte Público Metropolitano, 2026). A bus in the system runs about 60,000 kilometers per year, and the fleet-provision contracts run for 14 years, matching the useful life the tender framework assigns to an electric bus.¹⁷

At these values the energy saving alone, USD 0.425 per kilometer, amounts to USD 25,500 per bus-year, with a present value at 5.5 percent of USD 245,000 over the 14-year fleet contract, about twice the capital premium, and USD 145,000 over the 7-year operation contract, still above it. The maintenance saving adds to this margin; the main item on the other side is one battery replacement, since warranties run 5 to 7 years. Consistent with this arithmetic, DTPM reports that the 2023 and 2025 re-tenders reduced annual system payments by CLP 60 and 20 billion (USD 66 and 22 million), respectively. This matters for how we summarize the evaluation. Net benefits approximately equal gross benefits, so Table 8 is, to first order, the welfare effect itself. With a net fiscal cost near zero the marginal value of public funds diverges by construction, so we report instead the net social benefit per 10,000 electric bus-kilometers and, as an alternative summary, benefits per dollar of the gross capital premium.

7.4 Cross-validation and sensitivity

An independent benchmark exists for Santiago. Rizzi and De La Maza (2017) estimate marginal external costs for the diesel bus of the pre-electrification fleet, the same Euro III-era technology our design replaces. Expressed in dollars of December 2025 and at the official VVE, their estimates imply a diesel-bus externality of about 87 US cents per driven

¹⁷System kilometers were 390 to 415 million across an operating fleet of 6,400 to 6,900 buses in 2022–2025 (Directorio de Transporte Público Metropolitano, 2026). Fleet contracts run in 168 monthly quotas; the tender framework assigns electric buses a useful life of 14 years or 1.2 million kilometers (DTPM, 2025), in line with the 14-year bus life documented for United States transit agencies (Avenali et al., 2023). Operation contracts run for 7 years, renewable for up to 7 more.

bus-kilometer in air-pollution health costs, of which 53 cents operate through $\text{PM}_{2.5}$ and 34 cents through ozone precursors, and 5.1 cents in noise.¹⁸ Our noise saving is 13.6 cents per bus-kilometer with the 200-meter estimate and 4.7 cents with the 50-meter estimate, against their 5.1 cents. For pollution, the comparison depends on the mortality convention. At the VLY our valuation is 54 cents, the same as their $\text{PM}_{2.5}$ component; at the VVE it is 198 cents, the gap reflecting the annualization of an acute daily mortality response at the full statistical-life value. The two approaches are of the same order, and the largest gap sits where the mortality-valuation convention matters.

The comparison also shows what our valuation leaves out. We price the hourly change in $\text{PM}_{2.5}$ at roadside monitors, which captures primary emissions; secondary particles form over hours and days and over a wider area than a 750-meter buffer. Rizzi and De La Maza (2017) trace ambient concentrations back to precursor emissions and attribute 14 percent of ambient $\text{PM}_{2.5}$ to nitrogen oxides through secondary nitrate, and half of ambient ozone to the same source. Neither channel enters our estimate, so the NO_2 reduction we estimate has no counterpart in Table 8. Their ozone component is 15.3 cents per diesel bus-kilometer, or 34 cents at the VVE and 9 cents at the VLY; applied to the 162 million electric bus-kilometers of 2025, it would add about USD 55 and 15 million, some 15 and 12 percent of the respective totals. We leave it unvalued for two reasons. First, the authors present the ozone figure as an illustration, since it assumes that ozone formation in Santiago is limited by nitrogen oxides rather than by volatile organic compounds, and the local evidence is mixed (Seguel et al., 2012). Under the alternative regime a reduction in NO_x raises local ozone, so even the sign is unsettled. Second, their emission factors describe the pre-2015 fleet, while the diesel buses displaced after 2022 include Euro V and Euro VI vehicles with much lower NO_x emissions. Both points argue for omission, which makes the totals conservative on one further margin.

A second illustration uses the NO_2 coefficient directly. Baran et al. (2026) find that electric-vehicle adoption across US counties lowers NO_2 and the incidence of very low birth weight, and value each avoided case at a lifetime cost of USD 3.06 million. Dividing their

¹⁸Per driven bus-kilometer, in USD of 2015, their estimates are 23.3 cents of $\text{PM}_{2.5}$ -precursor health costs, 15.3 cents through ozone precursors, and 4.2 cents of noise, with congestion the largest component of a peak-hour total of USD 1.80. We inflate their 2015 pesos to December 2025 by the change in the UF (a factor of 1.55 in pesos, or 1.20 in dollars, since their exchange rate of CLP 707.34 differs from ours) and rescale their mortality-driven components from their VSL (USD 1.16 million of 2015, the mean of their uniform range) to the official VVE, a ratio of 1.87. Our per-kilometer figures divide the full-electrification benefits of Section 7.2 by 2025 total bus-kilometers (388.8 million). The two approaches also measure different objects, a composition effect at fixed service against the cost of an additional kilometer, so the comparison is of orders of magnitude.

birth-outcome effect by their NO_2 effect gives 0.47 to 0.69 fewer very-low-birth-weight births per 1,000 births per ppb. Our coefficient of 0.109 ppb per 10 electric bus-kilometers per hour implies a reduction of 1.7 ppb at full electrification and 0.7 ppb for the 2025 rollout. Applied to the 68,866 births of the Metropolitan Region in 2023 (Instituto Nacional de Estadísticas, 2026), the chain yields 23 to 33 avoided cases and USD 70 to 102 million per year, one-fifth to three-tenths of the 2025 total at the VVE. We do not add this figure either. Their estimate is the reduced-form effect of cleaner vehicles on births, attributed to NO_2 because that is the pollutant that moved in their data, so it overlaps with the $\text{PM}_{2.5}$ row. The lifetime cost is also a US cost-of-illness figure, not an official Chilean price.

The benchmark also clarifies what electrification cannot claim. Congestion and road damage, two of their externality components, are unchanged by fleet replacement: our operating-speed estimates are precise zeros, and electric buses are not lighter. We exclude both from the valuation. Finally, their passenger-kilometer comparison (4 cents for the bus against 41 to 42 cents for the car at peak) quantifies a margin our estimates do not contain: we find no short-run ridership response, but any rider eventually drawn from a car would add unpriced externality gains. Panel B of Table 8 collects the sensitivity of the 2025 total to the remaining choices: the carbon price, the noise row, and the exposed population; the mortality valuation is not a sensitivity but the dual VVE/VLY headline reported throughout. Across the rows, the 2025 total ranges from USD 318 to 389 million at the VVE and from USD 106 to 155 million at the VLY; the exposed population and the carbon price move it most.

8 Conclusion

We have used the staggered, contract-driven rollout of electric buses in Santiago to estimate what fleet electrification delivers on the ground. Predetermined contract transitions generated large increases in electric bus deployment, primarily through the replacement of conventional buses rather than an expansion of total service, providing a first source of variation in the timing of electrification. Complementary replacement regressions exploit local changes in electric and conventional bus activity while holding nearby total bus service fixed, isolating the effects of substituting electric for diesel operation. The resulting picture is consistent: replacing diesel with electric service lowers local air pollution and reduces ambient noise near bus corridors. By contrast, the contract transitions provide little evidence of detectable increases in ridership, while downstream effects on respiratory emergencies, school absenteeism,

and mortality are imprecisely estimated, as the modest size of the pollution reductions leads us to expect. Valued at official social prices, the environmental benefits amount to USD 128 to 362 million per year in 2025, depending on the mortality valuation, between one-tenth and three-tenths of the annual public subsidy to the system. The design of the tenders drove the incremental resource cost of the switch close to zero.

This is more than a local evaluation. The adopter is the state. The returns therefore do not hinge on the household adoption and utilization margins that limit green-transition instruments working through private choices, and they accrue directly to the people who live and travel along bus corridors. Against the menu of second-best transport policies, electrification is also unusual: it buys gains on several margins at once without the offsetting cost that most instruments carry on another. For the growing number of cities committed to zero-emission fleets, two lessons follow. The near-zero cost was engineered rather than automatic, since the tenders spread the capital premium of the electric fleet against its lower energy and maintenance costs. And the contracts did not raise ridership in the short run, so the estimates contain no gains from car trips diverted to quieter and cleaner buses; any such diversion over a longer horizon would add unpriced external gains.

Several things bound what these estimates establish. The estimates are package effects, capturing electrification as governments implement it, bundled with fleet renewal and new operations. They are also local and relative: displaced diesel buses may run elsewhere until they retire, so the estimates reflect the reallocation of bus emissions across space, and citywide general-equilibrium effects are absorbed by our time controls. Two channels travel less well than the headline. The carbon savings scale with the emissions intensity of the grid, which in Chile is decarbonizing quickly, so the same fleet would deliver different carbon gains elsewhere. And the Euro III diesel baseline replaced is dirtier than the incumbent fleets of many high-income cities, which raises the pollution and health returns relative to those settings. The outcomes we measure least well, downstream health and ambient noise, are also those where our data are coarsest, so that evidence is only suggestive.

As cities decarbonize urban travel, public transport is a margin where the state controls adoption directly and where the new technology replaces some of the dirtiest vehicles on the road. Santiago shows that the realized returns to this policy can be large, and that with the right contract design they can be obtained at little additional public cost.

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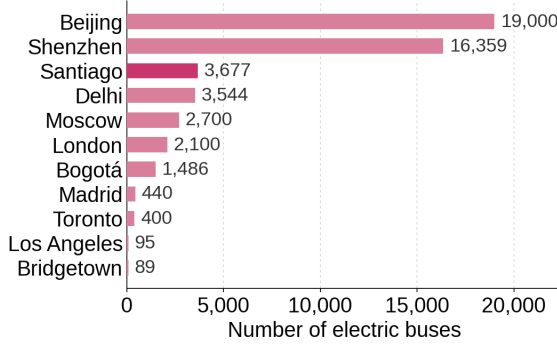
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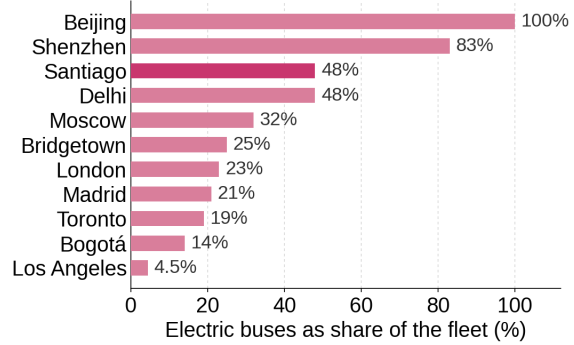
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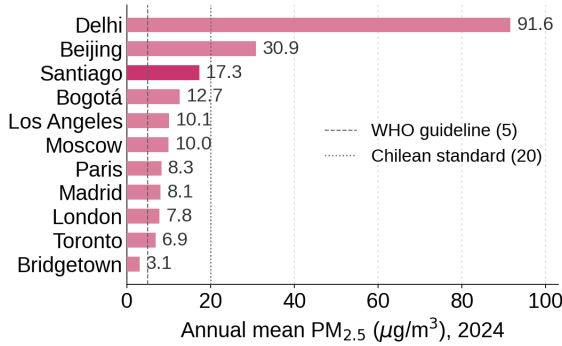
Figure 1: Santiago in comparative perspective



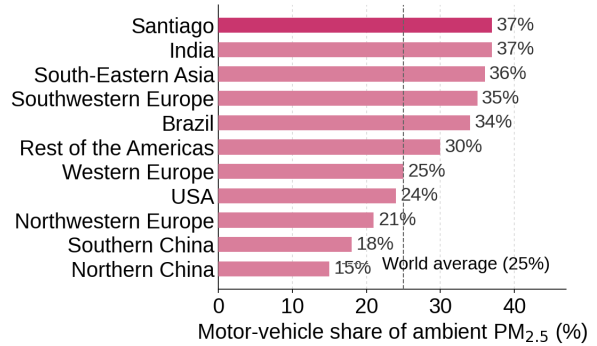
(a) Electric bus fleet, 2025



(b) Electric share of the bus fleet, 2025



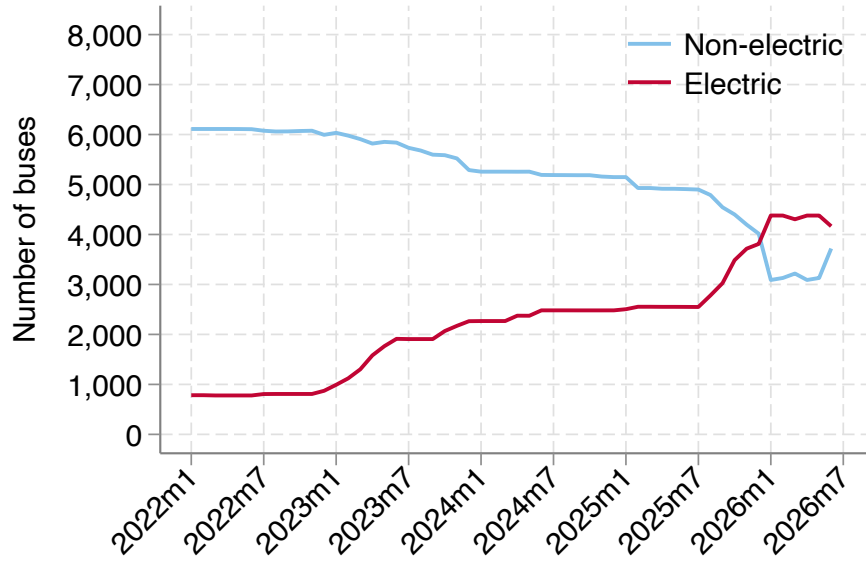
(c) Air pollution levels, 2024



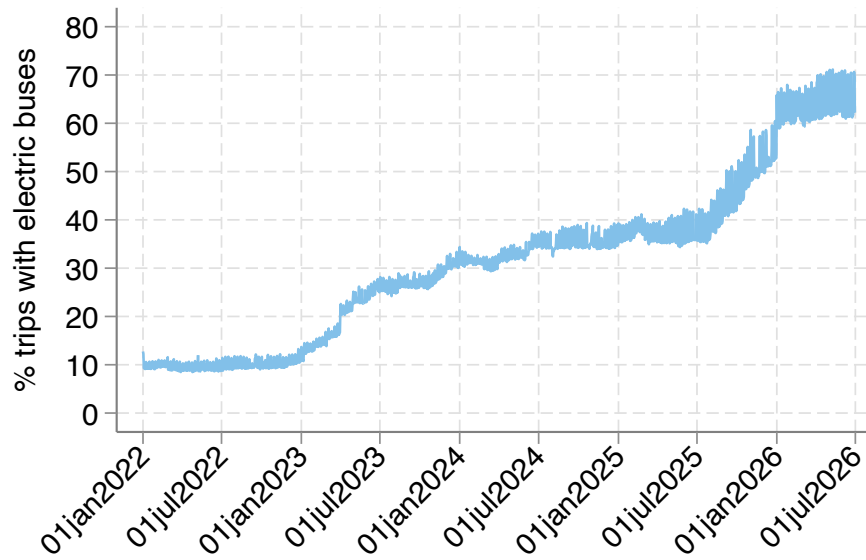
(d) Motor-vehicle share of ambient PM_{2.5}

Notes. Panels (a) and (b): electric buses in operation and electric buses as a share of the city bus fleet in 2025, from DTPM's Informe de Electromovilidad (DTPM, 2025), based on data from the Centro de Movilidad Sostenible; the Los Angeles figures are those reported by LA Metro as of September 2025 (Scauzillo, 2025). In panel (b), the Informe's chart transposes the Beijing and Shenzhen labels; we report the corrected assignment, consistent with the underlying Centro de Movilidad Sostenible data (Shenzhen's fleet has been fully electric since 2017). Panel (c): annual mean PM_{2.5} concentrations in 2024 (IQAir, 2025); the dashed and dotted lines mark the WHO annual guideline (5 µg/m³) and the Chilean annual standard (20 µg/m³). Panel (d): motor-vehicle (traffic) share of ambient PM_{2.5}; the Santiago estimate comes from 15 years of filter samples in central Santiago (Barraza et al., 2017), and the remaining bars are population-weighted regional averages of urban source-apportionment studies (Karagulian et al., 2015), whose traffic category includes exhaust and non-exhaust vehicle emissions; the dashed line marks the world average (25%).

Figure 2: Buses and electrification



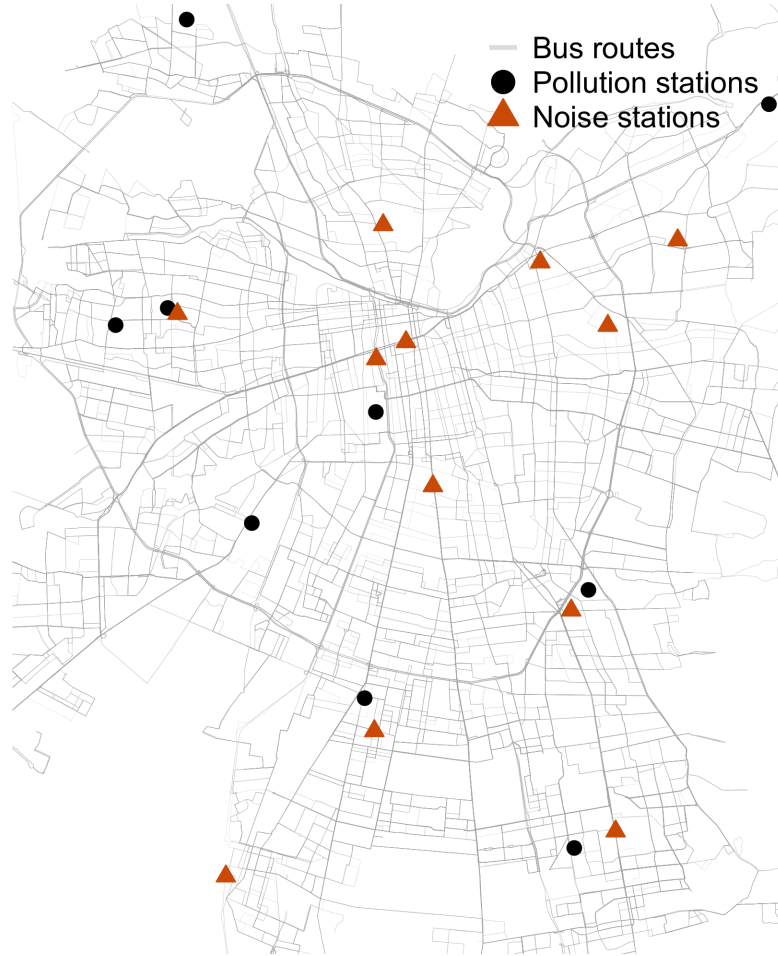
(a) Total number of buses by type



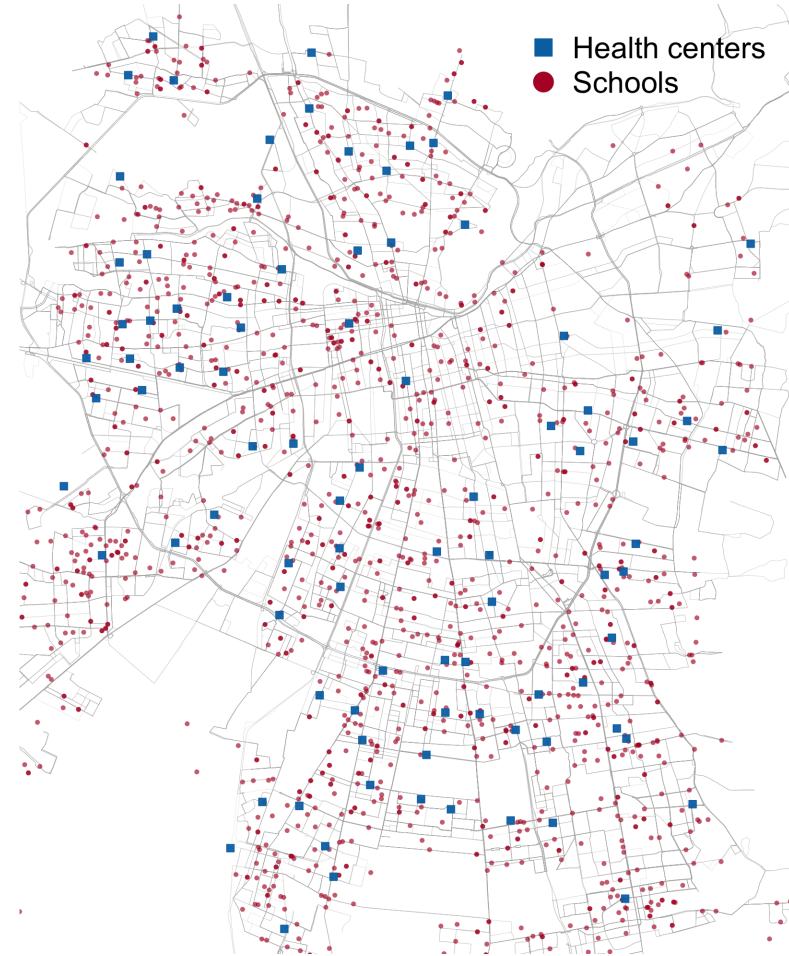
(b) Percentage of trips with electric buses

Notes. The figure documents the electrification of Santiago’s bus system from January 2022 through June 2026, using administrative records from the Ministry of Transportation. Panel (a) shows the composition of the operating fleet by propulsion technology. Panel (b) shows the daily share of bus trips operated by electric buses.

Figure 3: Bus routes, monitoring stations, health centers, and schools



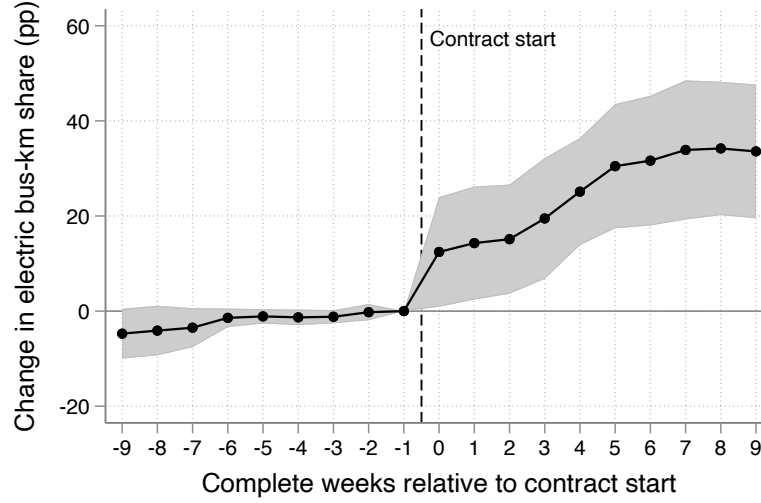
(a)



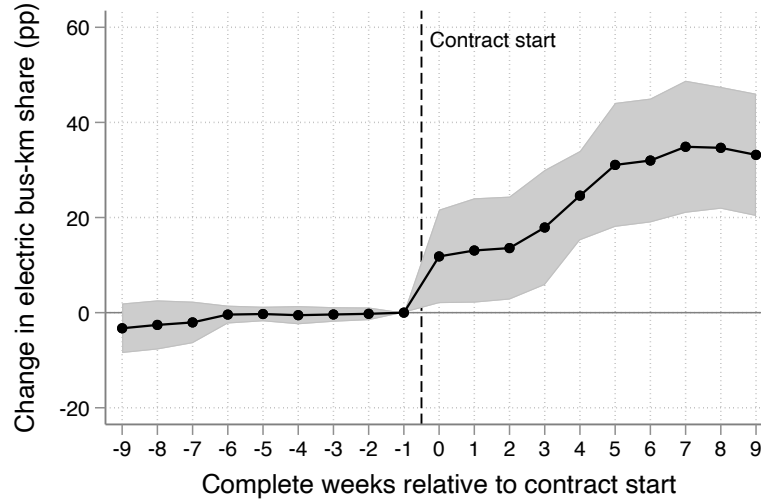
(b)

Notes. The figure shows a 2024 snapshot of the bus route network, together with the locations of air-quality monitoring stations, noise monitoring stations, health centers, and schools. Schools are restricted to those located within 5 km of the bus route network. Peripheral locations outside the displayed map extent are clipped.

Figure 4: Contract starts and electric-bus deployment



(a) Equal-weighted adoption path



(b) Stacked event-study estimates

Notes. Panel A shows the equal-weighted average change in the electric share of bus-kilometers across the 11 tendered service units, normalized to the final complete week before each contract starts. Panel B reports stacked event-study estimates in which each contract start defines a separate stack containing the affected route lineages and never-treated route lineages as controls. The sample covers complete event weeks -9 through 9 . The partial calendar week containing the formal contract start is omitted, so event week -1 is the final complete pre-contract week and event week 0 is the first complete post-contract week. The shaded areas report 95 percent confidence intervals. The regressions in panel B include stack-by-route-lineage and stack-by-calendar-week fixed effects, are weighted by pre-contract bus-kilometers, and use standard errors two-way clustered by contract and route lineage.

Table 1: Contract starts and electric-bus deployment

	Electric share (percentage points) (1)	Electric bus-km (% of pre-contract total) (2)	Non-electric bus-km (% of pre-contract total) (3)	Total bus-km (% of pre-contract total) (4)
Contract-start jump	12.17** (5.31)	11.79 (6.57)	-15.22*** (4.02)	-3.42 (4.31)
Weekly phase-in	3.02*** (0.93)	3.38*** (0.88)	-2.34** (0.77)	1.04*** (0.24)
Effect in event week 9	39.35*** (6.61)	42.19*** (6.90)	-36.24*** (7.00)	5.95* (3.10)
Pre-contract mean	16.07	16.06	83.94	100.00
Contracts	11	11	11	11
Treated route lineages	399	399	399	399
Observations	70,213	70,213	70,213	70,213
Stack-by-route-lineage fixed effects	Yes	Yes	Yes	Yes
Stack-by-calendar-week fixed effects	Yes	Yes	Yes	Yes
Weighted by pre-contract bus-kilometers	Yes	Yes	Yes	Yes

Notes. The table reports stacked event-study estimates of the effect of newly tendered service contracts on electric-bus deployment. Each of the 11 contract starts defines a separate stack containing the affected route lineages and never-treated route lineages as controls. The sample covers complete event weeks -9 through 9 ; the partial calendar week containing the contract start is omitted. The contract-start jump is the effect in event week 0 , the weekly phase-in is the additional linear change in subsequent weeks, and the event-week-9 effect equals the jump plus nine times the phase-in coefficient. Column 1 uses the electric share of bus-kilometers in percentage points. Columns 2–4 express electric, non-electric, and total bus-kilometers as percentages of pre-contract total bus-kilometers. All regressions include stack-by-route-lineage and stack-by-calendar-week fixed effects and are weighted by pre-contract bus-kilometers. Standard errors two-way clustered by contract and route lineage are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 2: Electric buses, operating speed, and passenger usage

	Service performance				Passenger response	
	Number of buses (1)	Bus-km (2)	Capacity-km (millions) (3)	Average speed (4)	Usage (5)	Ridership (6)
<i>Panel A: Workday peak hours</i>						
Share electric (km)	2.23*** (0.29)	19.34*** (5.98)	-0.013*** (0.002)	-0.02 (0.06)	11.68*** (1.58)	0.045** (0.020)
Contract: event week 9	-0.31 (0.67)	28.13* (12.65)	-0.002 (0.003)	-0.53** (0.19)	-1.99 (2.53)	-0.040 (0.037)
Avg. dependent variable	20.97	542	0.054	18.90	33.34	890
Observations, share specification	766,468	766,468	766,468	766,156	766,155	766,468
Routes	1,357	1,357	1,357	1,348	1,357	1,357
Observations, contract specification	66,055	66,055	66,055	65,979	66,055	66,055
Treated route lineages	382	382	382	381	382	382
<i>Panel B: Workday off-peak hours</i>						
Share electric (km)	4.44*** (0.47)	22.10** (9.12)	-0.017*** (0.004)	-0.04 (0.06)	10.12*** (1.48)	0.064*** (0.021)
Contract: event week 9	0.78 (0.59)	36.72** (12.82)	-0.001 (0.004)	-0.53** (0.20)	-0.68 (2.09)	-0.002 (0.031)
Avg. dependent variable	24.19	753	0.075	20.16	26.80	1,014
Observations, share specification	704,911	704,911	704,911	704,324	704,614	704,911
Routes	1,341	1,341	1,341	1,333	1,341	1,341
Observations, contract specification	60,968	60,968	60,968	60,846	60,968	60,968
Treated route lineages	374	374	374	373	374	374
<i>Panel C: Weekend/holiday all hours</i>						
Share electric (km)	4.20*** (0.65)	31.51*** (12.18)	-0.039*** (0.005)	0.10 (0.10)	9.96*** (1.55)	0.047*** (0.016)
Contract: event week 9	3.23*** (0.93)	91.24* (47.21)	-0.001 (0.015)	-0.43* (0.22)	-0.79 (1.93)	0.007 (0.022)
Avg. dependent variable	26.16	1,265	0.125	22.98	21.78	1,393
Observations, share specification	303,120	303,120	303,120	302,803	302,965	303,120
Routes	1,290	1,290	1,290	1,271	1,289	1,290
Observations, contract specification	55,055	55,055	55,055	54,945	55,055	55,055
Treated route lineages	330	330	330	328	330	330

Notes. The table reports route-level estimates from January 2022 through June 2026. “Share electric (km)” uses the share of route-day bus-kilometers operated by electric buses, with route and date fixed effects; standard errors are clustered by route. “Contract: event week 9” reports the reduced-form effect of contract starts using the stacked design in Figure 4, with stack-by-route-lineage and stack-by-calendar-week fixed effects and two-way clustered standard errors. All specifications except column 2 are weighted by average bus-km during the first 60 observed days in the case of realized-share specifications, while contract specifications use pre-contract bus-km. Capacity-km is measured in millions; usage is in percent; ridership is in logs, with the reported mean in levels. Panel C includes weekends and holidays. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 3: Electric bus adoption and hourly air pollution

Dependent variable:	PM _{2.5}	PM ₁₀	NO ₂	NO _x	CO
	(1)	(2)	(3)	(4)	(5)
Electric bus-km nearby, 10 km	-0.064 (0.031)	-0.322 (0.120)	-0.109 (0.053)	0.027 (0.276)	-0.0002 (0.0020)
Clustered <i>p</i> -value	0.088	0.037	0.086	0.924	0.908
Webb <i>p</i> -value	0.094	0.047	0.079	0.924	0.880
Electric signed-wind exposure, 10 km·m/s	-0.024 (0.081)	0.107 (0.240)	-0.008 (0.080)	0.139 (0.383)	-0.0018 (0.0038)
Clustered <i>p</i> -value	0.781	0.672	0.924	0.729	0.655
Webb <i>p</i> -value	0.837	0.672	0.916	0.753	0.657
Total bus-km nearby control	Yes	Yes	Yes	Yes	Yes
Total signed-wind exposure control	Yes	Yes	Yes	Yes	Yes
Station × date fixed effects	Yes	Yes	Yes	Yes	Yes
Station × hour-of-week fixed effects	Yes	Yes	Yes	Yes	Yes
Station × wind-sector fixed effects	Yes	Yes	Yes	Yes	Yes
Station × wind-speed-bin fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	212,187	215,094	205,934	205,940	200,688
Stations	7	7	7	7	7
Avg. dependent variable	22.97	64.99	21.67	50.16	0.69

Notes. The unit of observation is a pollution-monitoring-station by hour. The sample covers 1 January 2022 through 31 December 2025 and uses validated hourly pollution readings. Nearby bus activity is measured within 750 meters of each monitoring station. Electric and total bus-kilometers are measured in units of 10 km. Signed-wind exposure is the sum of nearby bus-kilometers multiplied by wind speed and by the cosine of the angle between the station-to-route bearing and the meteorological wind-from direction, and is measured in units of 10 km·m/s. Positive signed exposure corresponds to routes that are upwind of the monitor, so that wind carries emissions toward the station; negative exposure corresponds to routes for which emissions are carried away. Electric bus-kilometers, total bus-kilometers, electric signed-wind exposure, and total signed-wind exposure enter jointly in every specification, so the electric-bus coefficients capture changes in fleet composition holding total nearby bus activity fixed. All regressions include station-by-date, station-by-hour-of-week, station-by-eight-sector wind direction, and station-by-wind-speed-bin fixed effects. PM_{2.5} and PM₁₀ are measured in $\mu\text{g}/\text{m}^3$, and NO₂ and NO_x in parts per billion. Standard errors clustered by monitoring station are reported in parentheses. “Clustered *p*-value” reports inference based on these station-clustered standard errors. “Webb *p*-value” reports two-sided *p*-values from 9,999 restricted wild cluster bootstrap replications using Webb weights and clustering at the station level. The fixed effects are partialled out before implementing the wild cluster bootstrap; the resulting Frisch–Waugh–Lovell regressions reproduce the original point estimates numerically. No significance stars are reported.

Table 4: Electric bus adoption and ambient noise

	Distance from monitoring station		
	50 m (1)	100 m (2)	200 m (3)
<i>Panel A: Realized electrification</i>			
Electric bus-km, per 10 km	-3.74 (1.32)	-1.32 (0.69)	-0.60 (0.30)
Wild cluster bootstrap p -value	[0.072]	[0.106]	[0.093]
Full electrification at mean activity	-0.80	-0.89	-0.92
Mean total bus-km per hour	2.13	6.74	15.40
Observations	384,676	384,676	384,676
Stations	12	12	12
<i>Panel B: Contract-timing reduced form</i>			
Noise effect in event week 9	-0.137	-0.131	-0.002
Contract-jackknife p -value	[0.000]	[0.004]	[0.967]
First stage in event week 9, p.p.	2.59	2.29	1.82
First-stage jackknife p -value	[0.166]	[0.267]	[0.241]
Reduced-form observations	247,045	247,045	278,221
Contracts	7	7	8

Notes. Panel A reports hourly replacement specifications for January 2022 through December 2025. The dependent variable is hourly equivalent sound pressure in dB(A). Electric and total bus-kilometers are measured within the indicated distance from each monitoring station. Regressions include station-by-date and station-by-hour-of-week fixed effects and control for total nearby bus-kilometers. The reported coefficient is the change in dB(A) associated with replacing 10 nearby conventional bus-kilometers in an hour with 10 electric bus-kilometers, holding total bus activity fixed. “Full electrification at mean activity” evaluates this coefficient at mean total nearby bus-kilometers in the corresponding buffer. Parentheses report standard errors clustered by station; square brackets report restricted wild cluster bootstrap p -values using Webb weights and 9,999 replications. Panel B applies the contract-timing reduced form described in Section 3.3. Predetermined exposure is the share of nearby bus-kilometers in complete event weeks -9 through -1 supplied by each contract’s fixed route package; one unit corresponds to 10 percentage points of exposure. The table reports the implied effect in complete event week 9. The reduced form includes station-by-contract-by-hour-of-week and date-by-contract fixed effects and does not condition on contemporaneous total bus activity. The first stage reports the corresponding change in the local electric share, in percentage points. Square brackets report delete-one-contract jackknife p -values using a $t(G - 1)$ reference distribution. The number of contracts varies with the presence of positive predetermined exposure within each buffer.

Table 5: Emergency visits after high electric-bus adoption

	Dep. variable: Number of ER visits		Dep. variable: Respiratory composition	
	Respiratory	Non-respiratory	Respiratory share	Log respiratory/ non-respiratory
	(1)	(2)	(3)	(4)
0–52 weeks after high adoption	-0.009 (0.024)	-0.006 (0.016)	-0.002 (0.006)	-0.021 (0.035)
More than 52 weeks after high adoption	-0.055 (0.042)	-0.034 (0.033)	-0.005 (0.011)	-0.048 (0.064)
Mean dep. variable	233.7	501.0	0.318	-0.803
Health centers	69	69	69	69
Treated centers	68	68	68	68
Never-treated centers	1	1	1	1
Center-week observations	16,008	16,008	15,910	16,146
Health-center fixed effects	Yes	Yes	Yes	Yes
Week fixed effects	Yes	Yes	Yes	Yes

Notes. This table reports estimates from health-center-week regressions of emergency-room visits on indicators for the time elapsed since nearby bus service reached high electric adoption. High adoption is defined as an electric share of at least 20 percent among bus-kilometers within 2 km of the health center. Treated centers are required to contribute at least 49 pre-treatment weeks. The omitted category is the period before high adoption, including centers that do not reach the threshold during the sample. Columns 1 and 2 are estimated by PPML. Column 3 is estimated by OLS, weighted by total ER visits. Column 4 is estimated by OLS. All regressions include health-center and week fixed effects and control for total bus-kilometers within 2 km of the health center. Standard errors, clustered by health center, are reported in parentheses.

Table 6: Electric bus adoption and school absenteeism

Dependent variable:	Distance to bus routes			
	Absence rate, percentage points			
Buffer:	500m	1km	2km	5km
	(1)	(2)	(3)	(4)
Electric bus-km, 100k	-0.664 (0.641)	-0.300 (0.192)	-0.051 (0.055)	-0.004 (0.013)
Total bus-km, 100k	-0.665 (1.160)	-0.290 (0.374)	-0.109 (0.126)	-0.039 (0.031)
School fixed effects	Yes	Yes	Yes	Yes
Month-year fixed effects	Yes	Yes	Yes	Yes
Schools	1,315	1,315	1,315	1,315
Observations	57,149	57,149	57,149	57,149
Avg. dependent variable	13.02	13.02	13.02	13.02

Notes. The table reports school-month estimates of the relationship between electric-bus adoption and absenteeism among primary-school students. The sample covers March 2022 through June 2026 and includes regular primary-school students in grades 1–8. Student-month observations with 100 percent absenteeism are excluded before aggregating to the school-month level. The dependent variable is the school-month absence rate, defined as absent student-days divided by possible student-days and expressed in percentage points. Each column measures bus exposure within the buffer indicated in the column heading. Electric and total bus-kilometers are expressed in units of 100,000 km. Bus activity is constructed from bus-route-direction-day-time records and aggregated across buses and time periods before monthly spatial allocation. Route geometries are matched to the corresponding service periods using the available Ministry route shapefiles and official route crosswalks; for January–June 2026, the July 2026 route shapefile is used as the primary geometry, with the November 2025 shapefile as a fallback for unmatched routes. All specifications control for total bus-kilometers, so the coefficient on electric bus-kilometers captures changes in the composition of nearby bus service toward electric buses, holding total bus activity fixed. All regressions include school and month-year fixed effects. Standard errors clustered by school are reported in parentheses. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 7: Valuation inputs for the welfare synthesis

Parameter	Value	Source
<i>Prices (USD of December 2025)</i>		
Value of a statistical life (VVE)	USD 2.62 million (60,000 UF)	Ministerio de Desarrollo Social y Familia (2026)
Value per life-year (VLY)	USD 163,000	VVE annuitized, 5.5%, 40 years
VLY value per acute death	USD 706,000 (4.3 life-years)	Deryugina et al. (2019)
Cost per respiratory ER visit	USD 19.9 (CLP 18,130)	FONASA (2024)
Social price of carbon	USD 78.8/t CO ₂ (CLP 71,801)	Ministerio de Desarrollo Social y Familia (2026)
Social cost of carbon, range	USD 51; USD 190	Interagency Working Group on Social Cost of Greenhouse Gases (2021); U.S. Environmental Protection Agency (2023)
Social price of electricity, high tension	USD 0.131/kWh (CLP 119.8)	Ministerio de Desarrollo Social y Familia (2026)
Social price of diesel, bus	USD 0.883/liter (CLP 805)	Ministerio de Desarrollo Social y Familia (2026)
New urban bus, diesel; electric	USD 129,000; 247,000	Ministerio de Desarrollo Social y Familia (2026)
Annual kilometers per bus	≈60,000	Directorio de Transporte Público Metropolitano (2026)
Fleet contract; electric bus life	14 years; 14 years or 1.2 million km	DTPM (2025)
Noise depreciation index	0.40%/dB(A) [0.16–0.64]	Nelson (2008)
<i>Dose-response</i>		
Acute PM _{2.5} mortality, ages 65+	0.69 deaths/million per $\mu\text{g}/\text{m}^3$ -day	Deryugina et al. (2019)
Acute PM _{2.5} life-years lost, ages 65+	2.99/million per $\mu\text{g}/\text{m}^3$ -day	Deryugina et al. (2019)
Respiratory ER visits	+0.36% per $\mu\text{g}/\text{m}^3$ -day	Dardati et al. (2024)
<i>Quantities</i>		
Electric bus energy use	1.34 kWh/km	International Council on Clean Transportation (2024)
Diesel bus fuel economy	1.47 km/liter [1.39–1.55]	International Council on Clean Transportation (2022)
Grid emission factor (2023, official)	0.2384 t CO ₂ e/MWh	CEN (2024)
Bus activity within 750 m of PM _{2.5} monitors	154.3 bus-km per hour (42.5 electric)	own construction
Bus activity within 50 m; 200 m of noise monitors	2.13; 15.40 bus-km per hour	own construction
Population within 750 m of electrified routes	6,142,399 (aged 60+: 1,185,598)	own construction
Population within 200 m of electrified routes	5,217,626 (aged 60+: 1,020,450)	own construction
Dwellings within 50 m; 200 m of electrified routes	800,667; 1,987,201	own construction
Residential value within 50 m; 200 m	USD 105; 259 billion (mean sale, USD 131,000)	own construction
Baseline respiratory ER visits (2022, 2 km)	734,764	own construction
Annual bus-km, 2022–2025; 2026 H1	342.9 / 383.8 / 399.2 / 388.8; 187.1 million	own construction
Annual electric bus-km, 2022–2025; 2026 H1	45.0 / 96.1 / 137.8 / 161.9; 112.8 million	own construction
Pop.-weighted e-bus-km within 750 m, 2022–2025; 2026 H1	3.2 / 5.3 / 7.1 / 8.2; 5.8 million per year	own construction

Notes. All monetary values are US dollars of December 31, 2025, converted at USD 1 = CLP 911.18 and UF 1 = CLP 39,727.96 (USD 43.60). The high-tension electricity price applies to depot charging. The VVE follows the transfer function adopted by the Chilean investment-evaluation system; the noise depreciation index is the central value for road traffic noise, with the bracketed range spanning published meta-analytic estimates. The 2023 grid emission factor is the year-end value of the Coordinador Eléctrico Nacional; the 2022, 2024, and 2025 factors are provisional. Population and dwelling counts are area-weighted aggregates of 2024 census blocks within the stated distance of routes electrified by the end of 2025. Population-weighted electric bus-kilometers average, over all residents of the Metropolitan Region, the electric bus-kilometers operated on routes passing within 750 m of the resident’s census block; the 2026 entry is the six-month total. Bus activity at the monitors is the sample mean of the hourly regressions in Tables 3 and 4.

Table 8: The welfare effect of electrification

Panel A: citywide annual benefits over the rollout, USD million per year					
	2022	2023	2024	2025	2026 H1 [†]
Electric share of bus-km	13.1%	25.0%	34.5%	41.6%	60.3%
Electric bus-km, million	45.0	96.1	137.8	161.9	227.5
PM _{2.5} reduction, $\mu\text{g}/\text{m}^3$	0.13	0.25	0.34	0.41	0.60
Deaths avoided	39	74	102	123	178
Air pollution: mortality, at VVE	101	193	266	321	465
at VLY	27	52	72	87	126
Air pollution: morbidity	0.01	0.01	0.02	0.02	0.03
Noise, 200 m effect on the 200 m stock	6.9	13.2	18.2	22.0	31.8
alternative: 50 m effect on the 50 m stock	2.4	4.6	6.3	7.6	11.0
Carbon	5.3	11.4	16.3	19.2	27.0
Fare revenue	0	0	0	0	0
Total, mortality at VVE	113	218	301	362	524
Total, mortality at VLY	40	77	106	128	184
Panel B: 2025 total under alternative assumptions, USD million					
	Mortality at VVE		Mortality at VLY		
Central case	362		128		
Carbon price USD 51 per ton	356		121		
Carbon price USD 190 per ton	389		155		
No noise effect	340		106		
Noise: 50 m effect on the 50 m stock	348		114		
Exposed population within 200 m	318		116		

Notes. Panel A reports central-case citywide benefits by year, in millions of US dollars of December 2025.

[†]The 2026 column is the annualized running rate of the first half of the year (six-month totals scaled by 365/181). Each entry scales a replacement effect by that year’s electric share of bus-kilometers or electric bus-kilometers (Section 7.2). The PM_{2.5} reduction is the 750 m replacement coefficient of Table 3 at mean bus activity within 750 m of the monitors (154.3 bus-kilometers per hour), 0.99 $\mu\text{g}/\text{m}^3$ at full electrification, times the electric share. Mortality applies the acute response of Deryugina et al. (2019) to the 1.19 million residents aged 60 and older within 750 m of electrified routes, valued at the VVE (USD 2.62 million) or the VLY (USD 706,000). Morbidity applies the Dardati et al. (2024) response to baseline respiratory emergency visits at the FONASA tariff. Noise applies the -0.92 dB(A) effect at 200 m to the residential stock within 200 m (USD 259 billion) or, in the alternative row, the -0.80 dB(A) effect at 50 m to the stock within 50 m (USD 105 billion), capitalized at 0.40 percent per dB(A) and annualized at 5.5 percent; totals use the 200 m row. Carbon values the net saving of 15 t CO₂ per 10,000 electric bus-kilometers at USD 78.8 per ton. Fare revenue is zero because the contract design detects no ridership response (Section 4). The 2025 totals correspond to CLP 330 and 117 billion. Panel B varies one assumption at a time: the carbon price between USD 51 and 190 per ton; the noise row set to zero or to the 50 m alternative; and the exposed elderly population within 200 m (1.02 million) instead of 750 m.

ONLINE APPENDIX

Public Transport Electrification: Lessons from a Pioneer

Felipe González and Hugo E. Silva

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A Mortality Analysis

Because mortality accounts for most of the monetized air-pollution benefit in Section 7, we complement the pollution-based valuation with a direct analysis of administrative death records. The purpose of this exercise is not to replace the concentration–response approach used in the welfare calculation. Mortality is a rare outcome, the bus rollout changes only one component of ambient pollution, and the relevant effect implied by the pollution estimates is therefore difficult to identify directly. Instead, we ask whether the mortality data are consistent with a health effect of the magnitude used in the welfare calculation. The analysis uses the universe of individual death records maintained by the Departamento de Estadísticas e Información de Salud (DEIS) of the Chilean Ministry of Health. The records contain the date of death, age, comuna of residence, underlying cause of death, and external cause where applicable. We construct a weekly comuna panel and focus on deaths among individuals aged 65 and older from natural non-COVID causes, which most closely corresponds to the elderly population underlying the acute PM_{2.5} mortality response in ?. We use official DEIS records through 2023 and the corresponding 2024–2026 file for the subsequent period, retaining complete Monday–Sunday weeks through July 20, 2026.

We first relate mortality within Santiago to the predetermined contractual rollout. For each of the 11 contracts described in Section 2.3, let B_{cj} denote the population-weighted amount of weekly bus service in comuna c associated with the routes transferred under contract j , measured over the nine complete weeks preceding the contract start. We construct the contract-induced exposure measure

$$Z_{ct} = \sum_j B_{cj} 1\{t \geq T_j\}, \quad (9)$$

where T_j is the first complete week after the formal start of contract j . Because B_{cj} is fixed using service observed before the contract begins, Z_{ct} can be followed after 2025 without using realized bus operations in 2026. The contractual first stage is strong: as shown in Section 2.3, an additional 10,000 bus-kilometers of contract-induced exposure raises realized electric-bus exposure by approximately 5,650 bus-kilometers, while conventional bus-kilometers fall, so the variation generated by Z_{ct} is primarily a replacement of conventional service by electric service.

Regressions that use only variation across Santiago comunas produce a large negative association between contract exposure and elderly mortality. That association, however,

is not robust to comuna-specific mortality trends, and event-study estimates do not show mortality falling when electric operation actually phases in. In particular, the large negative coefficient is concentrated in contracts beginning in 2023 even though mortality does not decline over the first several months following those starts. This pattern raises a distinct identification concern from short-run anticipation or common shocks: comunas receiving more contract-induced electric-bus exposure may have been on different longer-run mortality trajectories from other Santiago comunas. We therefore construct an external counterfactual using comunas elsewhere in Chile, where the Santiago bus-contract rollout generates no treatment.

We define the donor pool using the 2024 Population Census and pre-treatment mortality only. The treated sample consists of the 42 Metropolitan Region comunas with positive exposure to Santiago bus service. Potential donors are comunas outside the Metropolitan Region that can be linked consistently between the Census and DEIS and satisfy common-support restrictions in population, age composition, and urban density. This leaves 169 eligible donor comunas. For each treated comuna we then construct nonnegative synthetic-control weights that sum to one. The matching information is restricted to 2015–2022 and places most weight on the relative annual path of age-65-plus natural non-COVID mortality, with additional weight on the pre-COVID mortality level, population and age composition, and pre-COVID mortality seasonality. Density is used to restrict the donor pool but is not targeted tightly in the weighting objective because the most densely populated Santiago comunas have no close analogue elsewhere in Chile. The regularization parameter governing weight concentration is selected using 2022 mortality as a pre-treatment holdout and the final weights are frozen before any mortality outcome after 2022 is used. The resulting controls are diffuse: the median effective number of donors is approximately 18, and the median largest donor weight is about 11 percent.

Appendix Figure A.7 assesses the resulting pre-treatment fit. The synthetic controls reproduce the aggregate mortality path closely over 2015–2019 and substantially reduce the discrepancy observed with simple nearest-neighbor controls during 2020–2022. Across the 42 treated comunas, the median root-mean-square error in the relative 2015–2022 mortality trajectory is 0.029; 37 comunas have an RMSE below 0.05. Appendix Table A.7 reports the corresponding construction and balance diagnostics. The remaining mismatch is concentrated in a small number of comunas and in the unusual mortality movements surrounding the COVID period. We therefore retain all 42 treated comunas in the baseline analysis and use the 37-comuna sample selected by the pre-specified RMSE threshold as a robustness

check. Importantly, both the donor pool and this fit criterion are determined entirely from information dated before the first contract starts.

To estimate the post-treatment effect, let m_{dt} denote the weekly mortality rate in donor comuna d and let w_{cd} denote the frozen synthetic weight that treated comuna c assigns to donor d . The synthetic counterfactual is

$$m_{ct}^S = \sum_d w_{cd} m_{dt}. \quad (10)$$

We express the synthetic rate on the treated comuna’s Census population denominator and stack the observed and synthetic outcomes. We then estimate Poisson pseudo-maximum-likelihood regressions with treated-comuna-by-side fixed effects and treated-comuna-by-calendar-week fixed effects. The treatment variable is Z_{ct} on the observed Santiago side of the pair and zero on its synthetic counterpart. The coefficient is therefore identified entirely by whether mortality in a Santiago comuna changes relative to its frozen national counterfactual as contract-induced electric-bus exposure increases. The preferred inference clusters by treated comuna and calendar week, and, as in the contract analysis, we omit the partial calendar week containing each formal contract start.

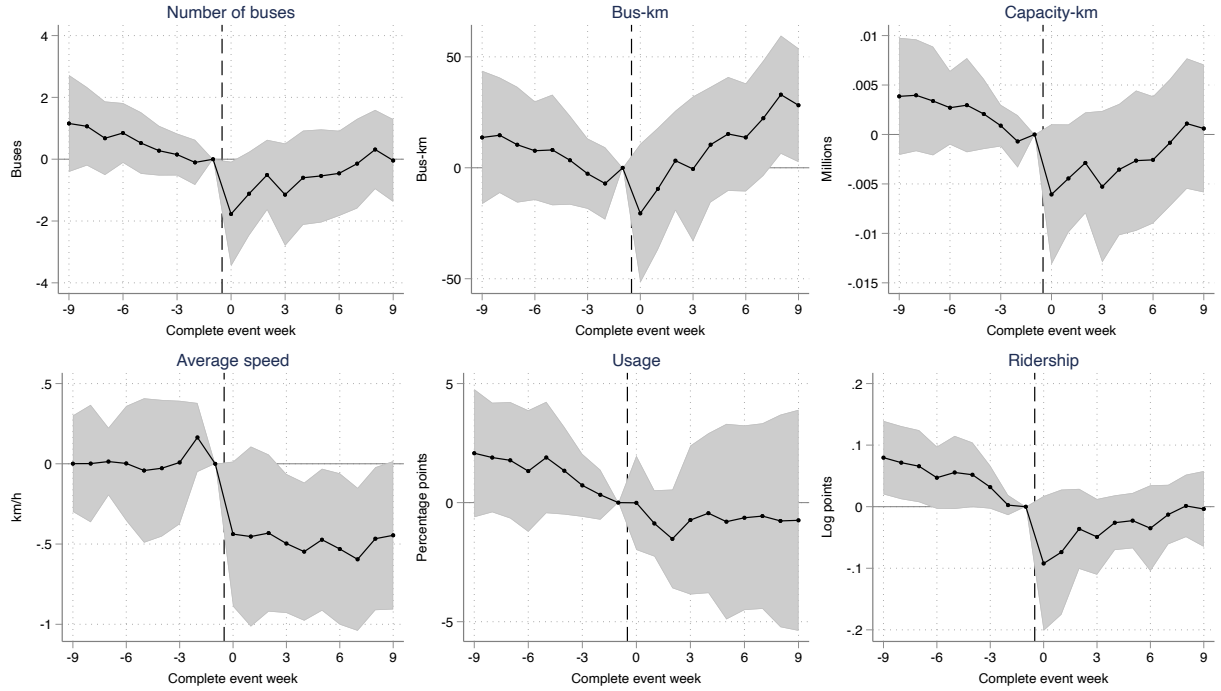
Appendix Table A.8 reports the results. For age-65-plus natural non-COVID mortality, the preferred estimate in the full 42-comuna sample is -0.000055 with a standard error of 0.000125 . The 95-percent confidence interval, $[-0.000307, 0.000196]$, comfortably includes zero. Restricting the sample to the 37 comunas with the strongest pre-treatment synthetic-control fit produces an estimate of -0.000075 with a standard error of 0.000122 . The conclusions are similar for all-cause elderly mortality, cardiorespiratory mortality, and all-age mortality, while non-transport external deaths provide a placebo outcome and likewise show no systematic response. Splitting contract exposure by tender wave also yields no precise mortality effect: the coefficient associated with the 2023 contracts is -0.000295 (s.e. 0.000217), whereas the coefficient associated with the 2025 contracts is 0.000149 (s.e. 0.000198). These results stand in contrast to the much larger negative association obtained when the comparison group is restricted to other Santiago comunas, indicating that much of that association reflects differential underlying mortality trajectories rather than the electrification shock itself.

Appendix Figure A.8 provides the corresponding dynamic evidence. We group complete event weeks into $[-9, -6]$, $[-5, -2]$, $[0, 2]$, $[3, 5]$, $[6, 9]$, $[10, 17]$, $[18, 26]$, and $[27, 38]$, with event

week -1 omitted. In the pooled sample of all 11 contracts, the pre-treatment coefficients are close to zero, and the post-treatment coefficients show no systematic decline in mortality as electric operation phases in. Joint tests yield $p = 0.923$ for the two pre-treatment bins, $p = 0.876$ for event weeks 0–9, and $p = 0.809$ for event weeks 10–38. The same conclusion holds in the 37-comuna good-fit sample. The lack of a detectable dynamic response is consistent with the static estimates: after conditioning on a counterfactual mortality trajectory constructed outside Santiago, the DEIS data do not identify a statistically precise mortality effect of the bus rollout.

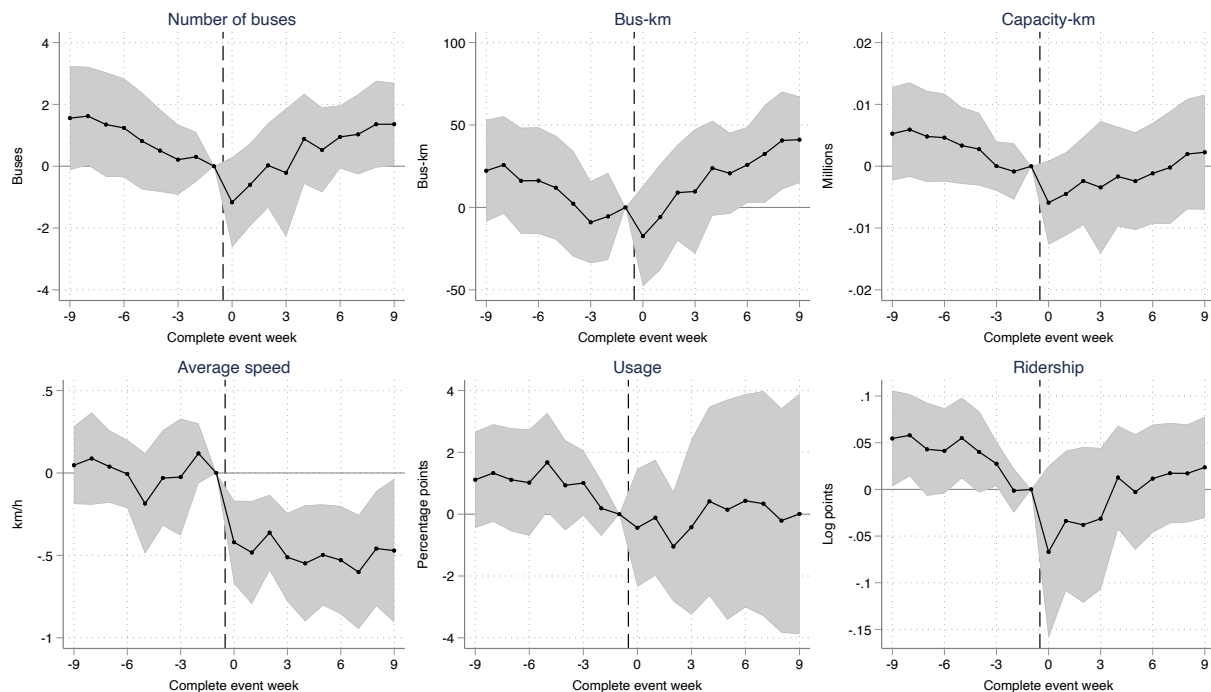
The direct mortality evidence nevertheless provides a useful check on the welfare calculation. Using the contract first stage only as a scale conversion, the preferred reduced-form coefficient corresponds to a point estimate of roughly -0.00010 per 10,000 realized electric bus-kilometers, far smaller than the large Santiago-only mortality association. More importantly, the confidence interval from the direct DEIS analysis comfortably contains a mortality reduction of the magnitude implied by the $PM_{2.5}$ calculation in Section 7. The direct estimates therefore do not provide an independent basis for monetizing mortality: they are too imprecise to distinguish such a modest effect from zero. But they also do not contradict the concentration–response approach. We consequently retain the pollution estimates combined with the external mortality response from ? as the basis for the welfare calculation, and interpret the DEIS exercise as an independent validation showing that the implied mortality benefit is consistent with, though not directly detectable in, Chilean mortality records.

Figure A.1: Contract starts and route outcomes: Workday peak hours



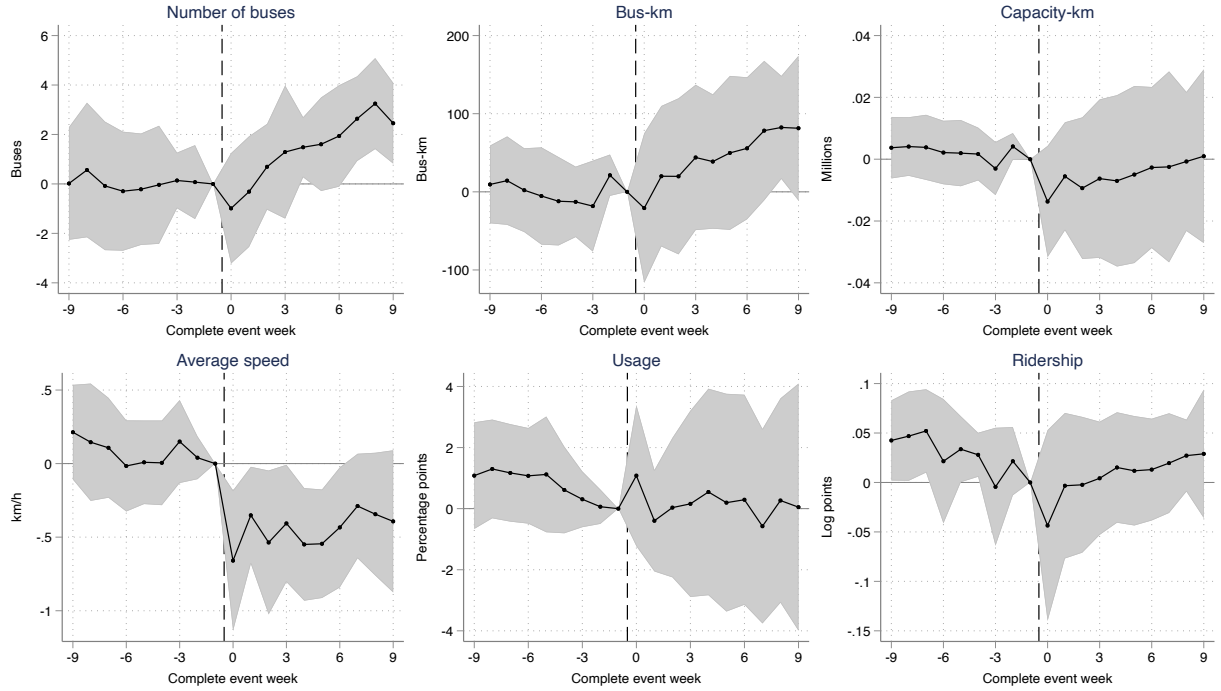
Notes. The figure reports stacked event-study estimates around the start of newly tendered service contracts for workday peak hours. Each contract defines a stack containing affected route lineages and never-treated route lineages as controls. Event week -1 , the final complete pre-contract week, is the omitted category; the partial calendar week containing the contract start is excluded. Regressions include stack-by-route-lineage and stack-by-calendar-week fixed effects and, except for bus-kilometers, are weighted by pre-contract bus-kilometers. Standard errors are two-way clustered by contract and route lineage. Shaded areas are 95 percent confidence intervals.

Figure A.2: Contract starts and route outcomes: Workday off-peak hours



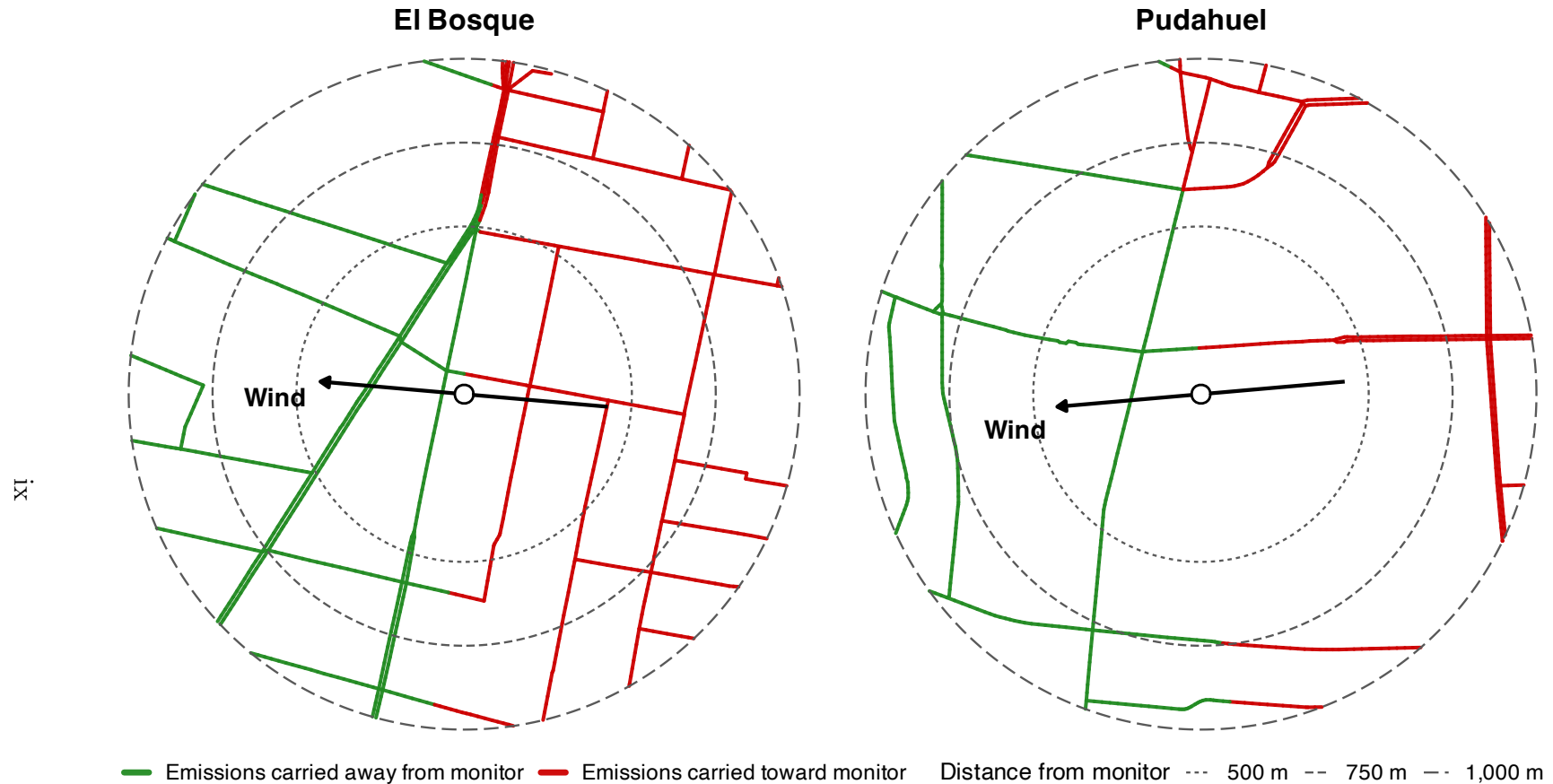
Notes. The figure reports stacked event-study estimates around the start of newly tendered service contracts for workday off-peak hours. Each contract defines a stack containing affected route lineages and never-treated route lineages as controls. Event week -1 , the final complete pre-contract week, is the omitted category; the partial calendar week containing the contract start is excluded. Regressions include stack-by-route-lineage and stack-by-calendar-week fixed effects and, except for bus-kilometers, are weighted by pre-contract bus-kilometers. Standard errors are two-way clustered by contract and route lineage. Shaded areas are 95 percent confidence intervals.

Figure A.3: Contract starts and route outcomes: Weekends and holidays



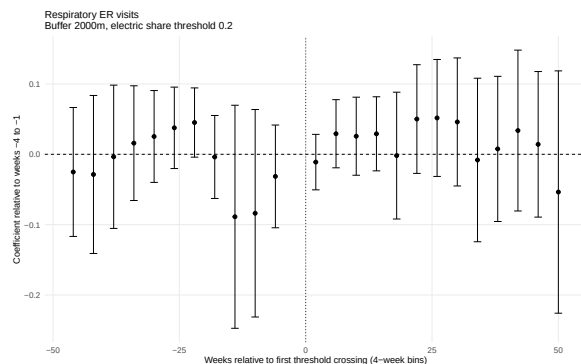
Notes. The figure reports stacked event-study estimates around the start of newly tendered service contracts for weekends and holidays. Each contract defines a stack containing affected route lineages and never-treated route lineages as controls. Event week -1 , the final complete pre-contract week, is the omitted category; the partial calendar week containing the contract start is excluded. Regressions include stack-by-route-lineage and stack-by-calendar-week fixed effects and, except for bus-kilometers, are weighted by pre-contract bus-kilometers. Standard errors are two-way clustered by contract and route lineage. Shaded areas are 95 percent confidence intervals.

Figure A.4: Wind direction and local bus-route exposure

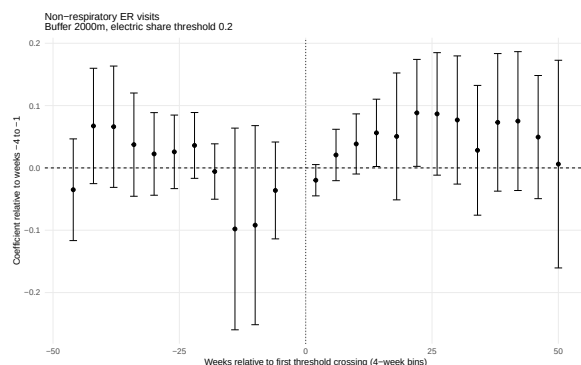


Notes. This figure illustrates the intuition of the wind-direction design using real bus-route geometries and the monitoring-station locations used in the analysis. Each panel is centered on one monitoring station. Concentric circles mark distances of 500 m, 750 m, and 1,000 m from the monitor. Red route segments denote routes whose emissions would be carried toward the monitoring station under the illustrated wind direction, while green route segments denote routes whose emissions would be carried away from the monitor. The wind direction shown in the figure is illustrative and chosen to clarify the identification strategy; the regression analysis uses observed wind data from SINCA. The empirical design compares pollution responses to electric-bus exposure in areas whose emissions are more likely to reach the monitor versus areas that are similarly close but, given wind direction, are less likely to affect measured pollution at that station.

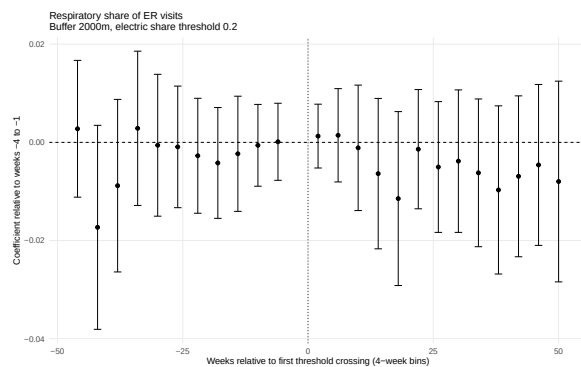
Figure A.5: Event study: ER visits around high electric-bus adoption



(a) Respiratory ER visits



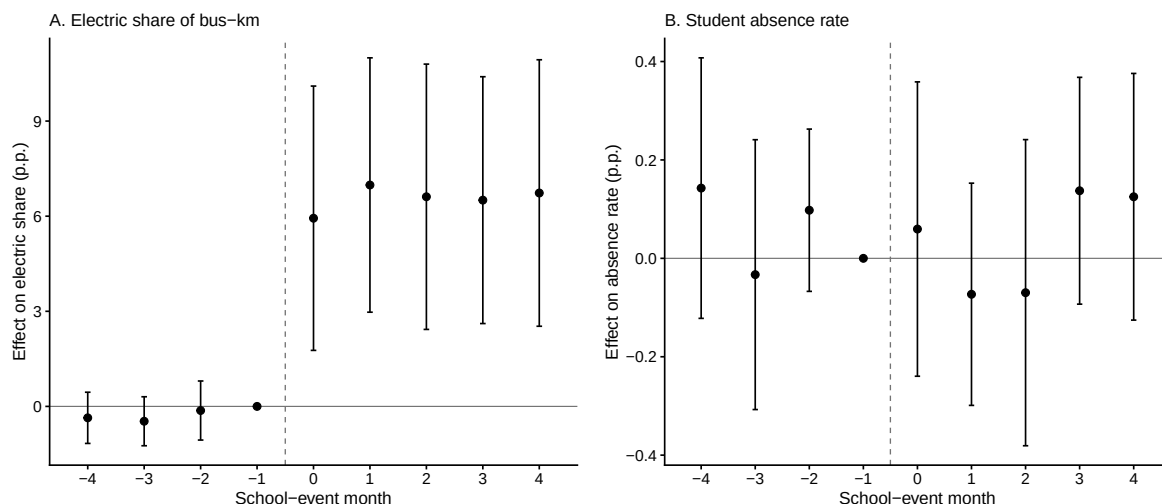
(b) Non-respiratory ER visits



(c) Respiratory share of ER visits

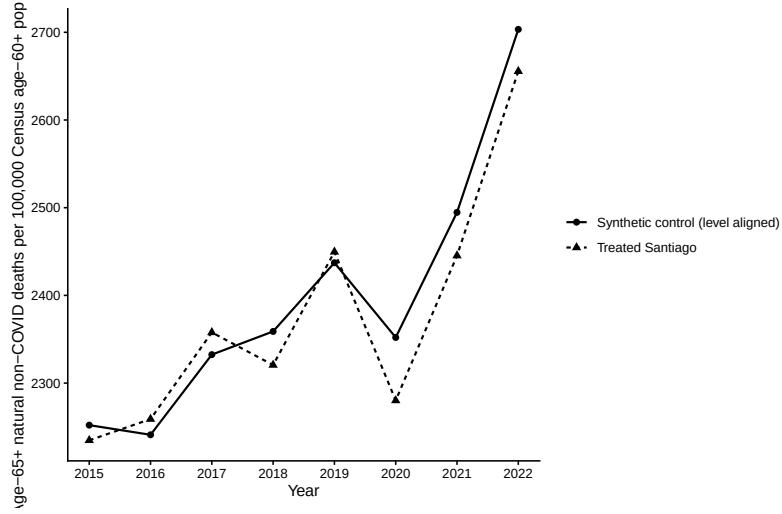
Notes. This figure plots event-study coefficients for emergency-room visits around the first week in which the electric share of bus-kilometers within 2 km of a health center reaches 20 percent. The omitted category is the four-week bin immediately before high adoption. Each coefficient corresponds to a four-week event-time bin. Panels A and B use $\log(1+y)$ outcomes for respiratory and non-respiratory ER visits, respectively. Panel C uses the respiratory share of ER visits. All regressions include health-center and week fixed effects and control for total bus-kilometers within 2 km of the health center. Vertical bars report 95 percent confidence intervals based on standard errors clustered by health center.

Figure A.6: Contract timing, electric-bus adoption, and school absenteeism

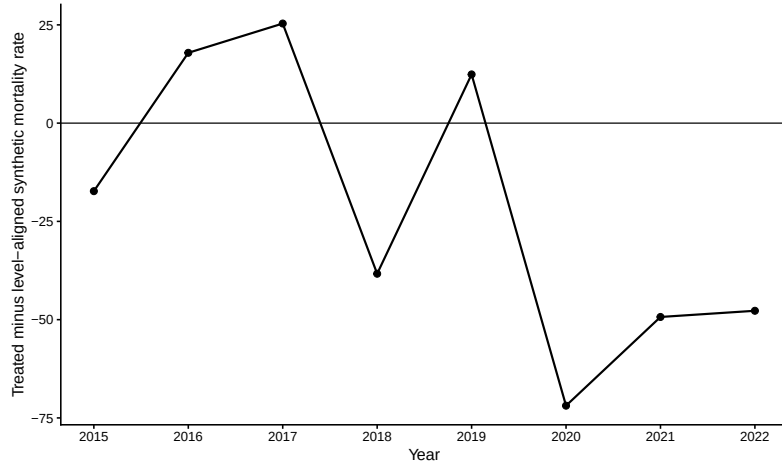


Notes. The figure reports event-study estimates from the contract-based design described in Appendix Table A.6. Panel A shows effects on the electric share of bus-kilometers within 2 km of each school, and Panel B shows effects on the school-month absence rate. Predetermined exposure is the share of nearby bus-kilometers supplied by each contract's fixed route package during the nine complete pre-contract weeks and is scaled so that one unit corresponds to a 10 percentage-point increase in exposure. Event time is measured in school months, excluding January, February, July, and December; if a contract begins during a school month, that month is omitted. The figure uses the balanced window from school-event month -4 through 4 , with event month -1 omitted. Each coefficient is the interaction between predetermined exposure and an indicator for the corresponding event month. Both specifications include school-by-contract and calendar-month-by-contract fixed effects. The first-stage specification is weighted by average nearby pre-contract bus-kilometers, while the absenteeism specification is unweighted. Points report coefficient estimates and bars report 95 percent confidence intervals constructed from delete-one-contract jackknife standard errors using a $t(10)$ reference distribution.

Figure A.7: Pre-treatment mortality fit of national synthetic controls



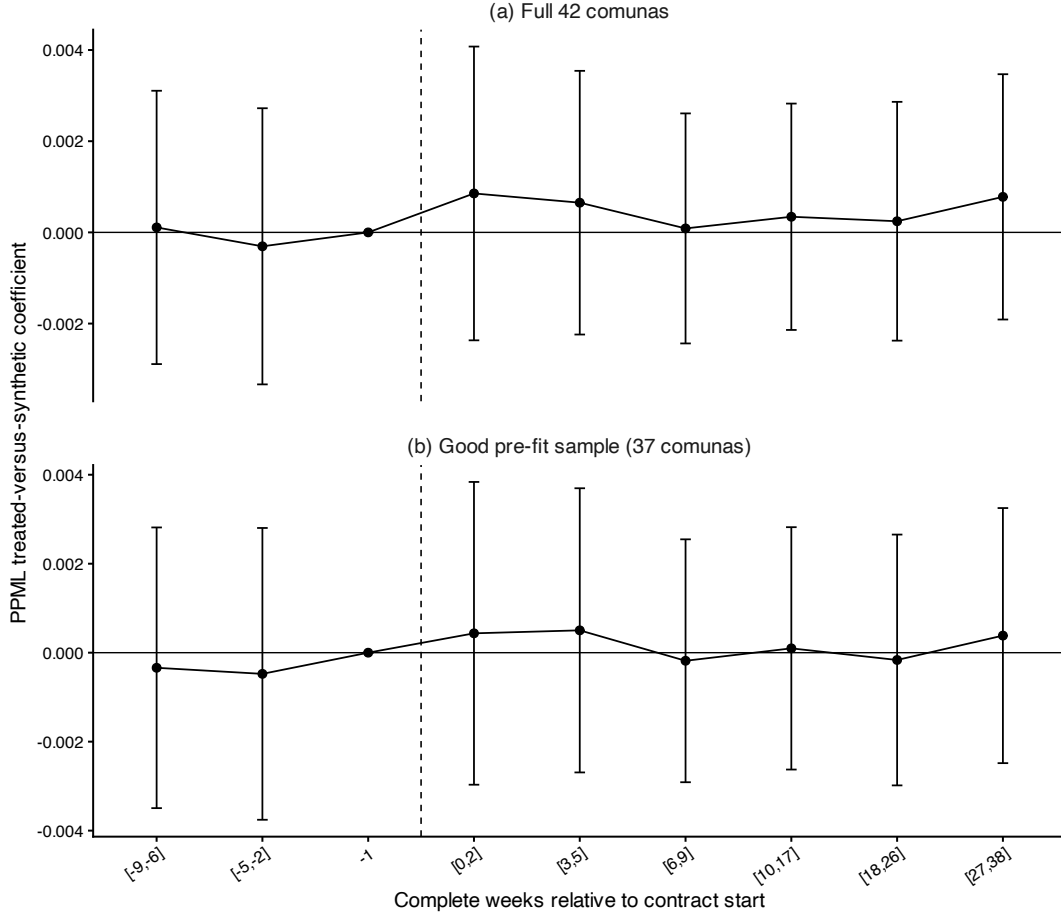
(a) Treated and synthetic mortality



(b) Treated minus synthetic mortality

Notes. The figure assesses the pre-treatment fit of the national synthetic controls used in the mortality analysis. The treated sample consists of the 42 Metropolitan Region comunas with positive exposure to Santiago bus service. For each treated comuna, the synthetic control is a convex combination of donor comunas outside the Metropolitan Region, with weights constructed using only information from 2015–2022 as described in Table A.7. The outcome is the annual rate of natural non-COVID deaths among individuals aged 65 and older, expressed per 100,000 residents aged 60 and older in the 2024 Census. For presentation, each synthetic mortality series is multiplicatively aligned to the corresponding treated comuna’s mean mortality rate over 2015–2019 before averaging across treated comunas. Panel (a) plots the average mortality rate in treated Santiago comunas and their synthetic controls. Panel (b) plots the corresponding treated-minus-synthetic difference. No mortality observation after 2022 is used to construct the donor pool, select the regularization parameter, or estimate the synthetic-control weights.

Figure A.8: Mortality around contract-driven electrification



Notes. The figure reports event-study estimates of mortality around the 11 contract starts that generate the contractual variation in electric-bus exposure. The dependent variable is the number of natural non-COVID deaths among individuals aged 65 and older. Each bus-served Santiago comuna is paired with the national synthetic control constructed using only pre-treatment information, as described in Table A.7. The treatment intensity for contract j in comuna c is the population-weighted amount of weekly bus service associated with the routes transferred under that contract, measured over the nine complete weeks preceding the contract start and expressed in units of 10,000 bus-kilometers. The calendar week containing the formal contract start is omitted because it combines operations under the old and new arrangements. Event week -1 , the final complete pre-contract week, is the omitted category. Coefficients are grouped into the event-time intervals shown on the horizontal axis. Panel (a) uses all 42 treated comunas. Panel (b) restricts the sample to the 37 comunas with a pre-treatment mortality-path RMSE no greater than 0.05. Regressions are estimated by PPML and include contract-by-treated-comuna-by-side and contract-by-treated-comuna-by-calendar-week fixed effects, as well as the exposure generated by other contracts. Standard errors are two-way clustered by treated comuna and calendar week, and vertical bars report 95 percent confidence intervals. In the full sample, tests that the two pre-treatment coefficients, the event-week 0–9 coefficients, and the event-week 10–38 coefficients are jointly zero yield p -values of 0.923, 0.876, and 0.809, respectively. The corresponding p -values in the good-pre-fit sample are 0.957, 0.869, and 0.794.

Table A.1: Robustness of hourly air-pollution estimates

Dependent variable:	PM _{2.5}	PM ₁₀	NO ₂	NO _x	CO
	(1)	(2)	(3)	(4)	(5)
<i>Panel A: Electric bus-km nearby, 10 km</i>					
Validated 2022–2025, wind sample	-0.064 (0.031)	-0.322 (0.120)	-0.109 (0.053)	0.027 (0.276)	-0.0002 (0.0020)
Clustered <i>p</i> -value	0.088	0.037	0.086	0.924	0.908
Webb <i>p</i> -value	0.094	0.047	0.079	0.924	0.880
Observations	212,187	215,094	205,934	205,940	200,688
Validated 2022–2025, no wind requirement	-0.002 (0.021)	-0.160 (0.072)	-0.041 (0.113)	0.376 (0.216)	0.0033 (0.0013)
Clustered <i>p</i> -value	0.929	0.069	0.727	0.133	0.041
Webb <i>p</i> -value	0.937	0.034	0.724	0.150	0.090
Observations	229,640	232,926	224,858	224,865	218,582
Including 2026 provisional data, wind sample	-0.014 (0.043)	-0.241 (0.142)	-0.145 (0.054)	0.020 (0.280)	0.0006 (0.0023)
Clustered <i>p</i> -value	0.750	0.140	0.037	0.945	0.799
Webb <i>p</i> -value	0.724	0.168	0.051	0.933	0.761
Observations	229,058	232,556	221,741	221,165	214,448
Including 2026 provisional data, no wind requirement	0.026 (0.037)	-0.150 (0.105)	-0.117 (0.114)	0.275 (0.259)	0.0034 (0.0021)
Clustered <i>p</i> -value	0.517	0.203	0.345	0.329	0.148
Webb <i>p</i> -value	0.492	0.159	0.354	0.312	0.122
Observations	247,258	251,197	241,369	240,751	232,996
<i>Panel B: Electric signed-wind exposure, 10 km·m/s</i>					
Validated 2022–2025	-0.024 (0.081)	0.107 (0.240)	-0.008 (0.080)	0.139 (0.383)	-0.0018 (0.0038)
Clustered <i>p</i> -value	0.781	0.672	0.924	0.729	0.655
Webb <i>p</i> -value	0.837	0.672	0.916	0.753	0.657
Including 2026 provisional data	-0.055 (0.091)	-0.032 (0.255)	-0.004 (0.068)	0.015 (0.465)	-0.0027 (0.0043)
Clustered <i>p</i> -value	0.567	0.905	0.953	0.976	0.556
Webb <i>p</i> -value	0.592	0.900	0.949	0.970	0.546
Buffer	750m	750m	750m	750m	750m
Stations	7	7	7	7	7

Notes. The table reports robustness checks for the hourly air-pollution estimates in Table 3. All specifications use a 750-meter buffer around each monitoring station. Panel A reports coefficients on electric bus-kilometers, controlling for total nearby bus-kilometers. Wind specifications additionally include electric and total signed-wind exposure and station-by-date, station-by-hour-of-week, station-by-wind-sector, and station-by-wind-speed-bin fixed effects. No-wind specifications omit the wind terms and wind fixed effects and therefore retain all station-hours with observed pollution. Panel B reports the electric signed-wind coefficient from the wind specifications. The 2022–2025 samples use validated pollution readings only; the 2026 extensions additionally use preliminary and not-yet-validated readings when validated values are unavailable. Standard errors clustered by monitoring station are in parentheses. Webb *p*-values are based on 9,999 restricted wild cluster bootstrap replications using Webb weights and station-level clustering. No significance stars are reported.

Table A.2: Contract timing, electric-bus adoption, and local air pollution

	First stage	Reduced-form pollution outcomes				
	Electric share (p.p.)	PM _{2.5}	PM ₁₀	NO ₂	NO _x	CO
	(1)	(2)	(3)	(4)	(5)	(6)
Contract-start jump $\times$ exposure	5.428 (1.871)	-1.279 (0.899)	-1.865 (1.758)	0.105 (0.331)	-3.095 (1.729)	-0.070 (0.019)
Weekly phase-in $\times$ exposure	-0.014 (0.079)	-0.118 (0.033)	-0.254 (0.060)	0.017 (0.035)	-0.378 (0.129)	-0.001 (0.001)
Effect in complete event week 9	5.306 (1.927)	-2.338 (1.129)	-4.149 (2.058)	0.254 (0.500)	-6.497 (2.554)	-0.083 (0.029)
Two-way clustered p -value, week 9	0.033	0.084	0.090	0.629	0.044	0.028
Contract-jackknife p -value, week 9	0.150	0.147	0.136	0.777	0.046	0.023
Pre-contract trend p -value	0.966	0.495	0.869	0.882	0.290	0.364
Late-preperiod placebo p -value	0.657	0.341	0.959	0.796	0.223	0.113
Pre-contract mean	21.8	25.55	72.22	21.38	48.86	0.663
Contracts	9	9	9	9	9	9
Monitoring stations	7	7	7	7	7	7
Observations	177,847	173,850	177,366	173,990	173,994	165,575
Station $\times$ contract $\times$ hour-of-week fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Date $\times$ contract fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Station $\times$ contract $\times$ wind controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes. The table exploits the official start dates of the tendered service contracts. For each monitoring station and contract, exposure is the share of bus-kilometers within 750 meters during complete event weeks -9 to -1 supplied by the contract's fixed pre-determined route package; one unit corresponds to a 10 percentage-point increase in this pre-contract exposure. Contracts with zero exposure at every monitoring station are excluded from the identifying sample. The partial calendar week containing contract start is omitted, and the window contains complete event weeks -9 through 9. The week-9 effect is the contract-start jump plus nine times the weekly phase-in coefficient. Parentheses report standard errors two-way clustered by monitoring station and contract. Contract-jackknife p -values use delete-one-contract inference with a $t(G - 1)$ reference distribution. Pre-contract trend and late-preperiod placebo p -values use the same contract-jackknife inference. Specifications use validated 2022–2025 pollution readings on the baseline wind sample and include station-by-contract-by-hour-of-week, date-by-contract, station-by-contract-by-wind-sector, and station-by-contract-by-wind-speed-bin fixed effects. Contemporaneous total bus-kilometers are not controlled for because service supply may itself respond to contract start. The reduced form captures the bundled effect of the new contracts and is not interpreted as an instrumental-variables estimate of electrification.

Table A.3: Robustness of the hourly ambient-noise estimates

	Distance from monitoring station		
	50 m (1)	100 m (2)	200 m (3)
<i>Panel A: Exact timestamp fixed effects</i>			
Electric bus-km, per 10 km	-2.57 (1.15)	-0.83 (0.56)	-0.34 (0.25)
Full electrification at mean activity	-0.55	-0.56	-0.52
Observations	384,676	384,676	384,676
Stations	12	12	12
<i>Panel B: Contract-design diagnostics</i>			
Pre-contract trend coefficient	-0.0018	-0.0017	0.0003
Contract-jackknife <i>p</i> -value	[0.783]	[0.808]	[0.913]
Late-preperiod placebo effect	-0.005	-0.007	0.017
Contract-jackknife <i>p</i> -value	[0.870]	[0.809]	[0.390]
<i>Panel C: Leave-one-contract-out stability</i>			
Baseline week-9 reduced form	-0.137	-0.131	-0.002
Minimum leave-one-out estimate	-0.150	-0.158	-0.032
Maximum leave-one-out estimate	-0.128	-0.119	0.020
Identifying contracts	7	7	8
Stations with positive bus exposure	7	7	10

Notes. Panel A augments the preferred hourly replacement specification with exact date-by-hour fixed effects, in addition to station-by-date and station-by-hour-of-week fixed effects. The dependent variable is hourly equivalent sound pressure in dB(A). Electric and total nearby bus-kilometers are measured within the indicated buffer and enter in units of 10 km. The reported coefficient is the change in dB(A) associated with replacing 10 conventional bus-kilometers with 10 electric bus-kilometers while holding total bus activity fixed. Parentheses report standard errors clustered by station. “Full electrification at mean activity” evaluates the coefficient at mean total bus-kilometers in the corresponding buffer. Panel B reports diagnostics for the contract-timing reduced form in Panel B of Table 4. The pre-contract trend test interacts predetermined contract exposure with a linear event-time trend over complete event weeks -9 through -1 . The placebo assigns a false contract start in event week -4 and reports its implied effect in event week -1 . Square brackets report delete-one-contract jackknife *p*-values. Panel C reports the range of event-week-9 reduced-form estimates obtained by omitting each identifying contract in turn. “Identifying contracts” counts contract episodes with positive predetermined exposure to at least one monitoring station within the indicated buffer. The final row reports the number of monitoring stations that ever have positive nearby bus exposure over the estimation sample.

Table A.4: Contract timing, electric-bus adoption, and emergency visits

	First stage	Reduced-form health outcomes			
	Electric share (p.p.) (1)	Respiratory ER visits (2)	Non-respiratory ER visits (3)	Respiratory share (4)	Log respiratory/ non-respiratory (5)
Effect in complete event week 9	6.826 (1.464)	0.016 (0.005)	0.003 (0.011)	0.002 (0.003)	0.003 (0.011)
Contract-jackknife p -value	0.001	0.002	0.807	0.540	0.820
Pre-contract trend p -value	0.264	0.624	0.158	0.558	0.424
Late-preperiod placebo p -value	0.161	0.202	0.320	0.316	0.113
Pre-contract mean	27.6	221.9	503.9	0.306	-0.848
Contracts	11	11	11	11	11
Health centers	95	95	95	95	95
Observations	19,855	19,665	19,665	19,651	19,855
Center $\times$ contract fixed effects	Yes	Yes	Yes	Yes	Yes
Week $\times$ contract fixed effects	Yes	Yes	Yes	Yes	Yes

Notes. This table reports a contract-timing first stage and reduced-form health estimates using the fixed route packages and official start dates of the 11 tendered service-unit contracts. For each health center and contract, predetermined exposure is the share of bus-kilometers within 2 km of the center during complete event weeks -9 to -1 that belongs to the contract's fixed route package. Regressors are scaled so that one unit corresponds to a 10 percentage-point increase in predetermined contract exposure. The partial calendar week containing the contract start is omitted, and the estimation window contains complete event weeks -9 through 9. The reported event-week-9 effect is the implied effect from a specification allowing for a discrete change at contract implementation followed by a linear phase-in. Column 1 is estimated by OLS and weighted by average nearby pre-contract bus-kilometers. Columns 2 and 3 are estimated by PPML. Column 4 is estimated by OLS and weighted by total ER visits. Column 5 is estimated by OLS. All specifications include health-center-by-contract and week-by-contract fixed effects and do not control for contemporaneous total bus-kilometers, which may itself respond to the contract. Standard errors in parentheses are two-way clustered by health center and contract. Because identification uses only 11 contract timing shocks, the next row reports delete-one-contract jackknife p -values using a $t(10)$ reference distribution. The pre-contract trend test interacts predetermined exposure with linear event time over weeks -9 to -1 . The late-preperiod placebo assigns a pseudo treatment beginning in event week -4 and evaluates its implied effect at week -1 ; reported diagnostic p -values use the same delete-one-contract jackknife.

Table A.5: Emergency-visit estimates across health-center samples

Sample	Centers; treated/never	Respiratory ER visits		Non-respiratory ER visits		Log respiratory/non-respiratory	
		0–52 weeks	>52 weeks	0–52 weeks	>52 weeks	0–52 weeks	>52 weeks
All centers	69; 68/1	-0.009 (0.024)	-0.055 (0.042)	-0.006 (0.016)	-0.034 (0.033)	-0.021 (0.035)	-0.048 (0.064)
SAPU only	57; 56/1	-0.016 (0.029)	-0.078 (0.050)	-0.009 (0.020)	-0.043 (0.042)	-0.032 (0.041)	-0.070 (0.075)
Excluding top 5 volume centers	65; 64/1	0.002 (0.025)	-0.048 (0.045)	-0.010 (0.018)	-0.051 (0.036)	-0.014 (0.037)	-0.036 (0.067)
Excluding top 10 volume centers	61; 60/1	-0.005 (0.028)	-0.070 (0.049)	-0.009 (0.019)	-0.062 (0.040)	-0.023 (0.040)	-0.052 (0.072)
Excluding low-volume or unstable centers	63; 62/1	-0.009 (0.024)	-0.042 (0.041)	-0.006 (0.017)	-0.026 (0.033)	-0.020 (0.037)	-0.046 (0.068)
SAPU, excluding top 10 or unstable centers	54; 53/1	-0.005 (0.030)	-0.070 (0.053)	-0.013 (0.022)	-0.062 (0.045)	-0.024 (0.043)	-0.057 (0.078)

Notes. Each row reports the preferred event-window specification for a different health-center sample. High electric-bus adoption is the first week in which the electric share of bus-kilometers within 2 km of a health center reaches 20 percent. Treated centers must contribute at least 49 pre-treatment weeks. The first two outcome groups are estimated by PPML; the log respiratory/non-respiratory outcome is estimated by OLS. All regressions include health-center and week fixed effects and control for total bus-kilometers within 2 km. Standard errors clustered by health center are reported in parentheses. Volume rankings are calculated among all 95 health centers. Low-volume centers are below the 10th percentile of average weekly ER visits, and unstable centers have an absolute change above 50 percent between the first and last 52 weeks. The rows excluding only unstable centers and excluding top-10-volume or unstable centers are omitted because they reproduce exactly the all-center and excluding-top-10 samples, respectively.

Table A.6: Contract timing, electric-bus adoption, and school absenteeism

	First stage	Reduced form
	Electric share (p.p.)	Absence rate (p.p.)
	(1)	(2)
First post-contract school-month jump $\times$ exposure	6.570 (1.658)	-0.084 (0.085)
School-month phase-in $\times$ exposure	0.111 (0.266)	0.034 (0.026)
Effect in school-event month 4	7.016 (1.802)	0.052 (0.055)
Contract-jackknife p -value	0.003	0.296
Pre-contract trend p -value	0.220	0.453
Late-preperiod placebo p -value	0.268	0.633
Pre-contract mean	25.5	13.18
Contracts	11	11
Schools	1,291	1,291
Observations	125,652	125,677
School $\times$ contract fixed effects	Yes	Yes
Month $\times$ contract fixed effects	Yes	Yes

Notes. The table reports a contract-timing first stage and reduced-form estimate of the effect of bus-service contracts on school absenteeism. The design uses the fixed route packages and official start dates of 11 tendered service-unit contracts. For each school and contract, predetermined exposure is the share of bus-kilometers within 2 km of the school during the nine complete pre-contract weeks -9 to -1 supplied by the contract's fixed route package. Exposure is scaled so that one unit corresponds to a 10 percentage-point increase in this predetermined share. Event time is measured in school months: March–June and August–November; January, February, July, and December are omitted as school-holiday months. If a contract begins during a school month, that month is omitted. The balanced estimation window covers school-event months -4 through 4. The event-month-4 effect equals the first post-contract school-month jump plus four times the school-month phase-in coefficient. Parentheses report conventional standard errors two-way clustered by school and contract. Because there are only 11 contract shocks, the table also reports delete-one-contract jackknife p -values for the event-month-4 effect using a $t(10)$ reference distribution. The pre-contract trend test interacts predetermined exposure with linear school-event time over months -4 through -1 . The late-preperiod placebo applies the same jump-plus-phase structure beginning in event month -2 and reports its implied effect at event month -1 ; reported p -values for both diagnostics use the same delete-one-contract jackknife. Column 1 is estimated by OLS and weighted by average nearby pre-contract bus-kilometers. Column 2 is estimated by unweighted OLS. Both specifications include school-by-contract and calendar-month-by-contract fixed effects. The reduced-form specification does not control for contemporaneous total bus-kilometers, which may itself respond to contract implementation. The contract design is interpreted as a complementary reduced-form institutional robustness exercise rather than as an estimate of the effect of electric propulsion alone.

Table A.7: Construction and pre-treatment fit of national synthetic controls

<i>Panel A: Sample construction</i>			
Treated Santiago comunas	42		
Potential donor comunas outside the Metropolitan Region	282		
Eligible donor comunas	169		
Donor comunas receiving positive weight	144		
Pre-treatment mortality period	2015–2022		
Holdout year for regularization	2022		
<i>Panel B: Synthetic-control weights and pre-treatment fit</i>			
Selected regularization parameter, λ	1.000		
2022 holdout RMSE	0.104		
Median effective number of donors	17.6		
Median largest donor weight	0.108		
Maximum largest donor weight	0.324		
Median 2015–2022 mortality-path RMSE	0.029		
Maximum 2015–2022 mortality-path RMSE	0.094		
Comunas with mortality-path RMSE ≤ 0.05	37		
Median absolute 2022 mortality-rate gap	38.8		
<i>Panel C: Covariate balance</i>			
	Treated mean	Synthetic mean	Std. difference
Population, 2024	160,381	112,367	0.400
Share age 60 and older	0.192	0.195	-0.073
Female share	0.517	0.517	-0.033
Population density, persons/km ²	9,354	4,161	1.070
Pre-COVID mortality rate	2,324.3	2,516.8	-0.378
Winter mortality share	0.301	0.292	0.693
Summer mortality share	0.218	0.223	-0.452
Seasonal amplitude	0.042	0.044	-0.303

Notes. The treated sample consists of the 42 Metropolitan Region comunas with positive exposure to Santiago bus service. Potential donors are comunas outside the Metropolitan Region. Eligibility is determined using the 2024 Population Census and pre-treatment mortality information and is fixed before any post-2022 mortality outcome is used. Synthetic-control weights are nonnegative and sum to one separately for each treated comuna. The matching objective gives greatest weight to the relative 2015–2022 trajectory of age-65-plus natural non-COVID mortality and additionally uses the pre-COVID mortality level, population and demographic characteristics, and pre-COVID mortality seasonality. Population density is used to screen the donor pool but is not targeted directly in the weighting objective because the density of central Santiago lies outside the support of most other Chilean comunas. The regularization parameter is selected using 2022 mortality as a pre-treatment holdout; among candidate values within 5 percent of the minimum holdout RMSE, we select the largest value to favor more diffuse weights. The effective number of donors is $1/\sum_d w_{cd}^2$. Mortality-path RMSE is calculated from each comuna’s annual mortality rate relative to its own 2015–2019 mean. The 2022 mortality-rate gap is the absolute difference between treated and level-aligned synthetic mortality, measured as age-65-plus natural non-COVID deaths per 100,000 residents aged 60 and older in the 2024 Census. The good-fit sample used below consists of the 37 treated comunas with a pre-treatment mortality-path RMSE no greater than 0.05. In Panel C, the synthetic mean averages the weighted donor characteristic across treated comunas, and the standardized difference is the mean treated-minus-synthetic difference divided by the standard deviation of the characteristic across treated comunas.

Table A.8: Electric-bus contracts and mortality: national synthetic controls

Sample:	All contracts		2023 contracts		2025 contracts	
	Full (1)	Good pre-fit (2)	Full (3)	Good pre-fit (4)	Full (5)	Good pre-fit (6)
<i>Panel A: Mortality outcomes</i>						
Natural non-COVID deaths, age 65+	-0.000055 (0.000125)	-0.000075 (0.000122)	-0.000295 (0.000217)	-0.000321 (0.000219)	0.000149 (0.000198)	0.000139 (0.000197)
All deaths, age 65+	-0.000073 (0.000119)	-0.000080 (0.000119)	-0.000313 (0.000202)	-0.000317 (0.000206)	0.000146 (0.000198)	0.000139 (0.000198)
Cardiovascular or respiratory deaths, age 65+	-0.000105 (0.000171)	-0.000123 (0.000168)	–	–	–	–
All-cause deaths, all ages	-0.000033 (0.000103)	-0.000021 (0.000103)	–	–	–	–
Non-transport external deaths	-0.000197 (0.000415)	-0.000163 (0.000438)	0.000691 (0.000804)	0.000628 (0.000855)	-0.001003 (0.000613)	-0.000875 (0.000594)
<i>Panel B: Specification and sample</i>						
Treated comunas	42	37	42	37	42	37
Treated-comuna-by-side fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Treated-comuna-by-week fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Two-way clustered standard errors	Yes	Yes	Yes	Yes	Yes	Yes
Stacked observations	19,068	16,798	19,068	16,798	19,068	16,798

Notes. The table reports Poisson pseudo-maximum likelihood estimates comparing observed mortality in each bus-served Santiago comuna with mortality in its frozen national synthetic control. For treated comuna c , synthetic mortality in week t is the weighted average of mortality rates among donor comunas, using the weights summarized in Table A.7. The weights are estimated using only information from 2015–2022 and are not re-estimated using post-treatment mortality. Synthetic mortality rates are expressed on the treated comuna’s fixed 2024 Census population denominator before the treated and synthetic observations are stacked. The treatment variable is contract-induced bus exposure, measured in units of 10,000 bus-kilometers using the fixed pre-contract exposure associated with contracts that have begun operation. The treatment variable takes its observed value on the Santiago side of each pair and zero on the synthetic side. All regressions include treated-comuna-by-side and treated-comuna-by-calendar-week fixed effects, so identification comes from changes in mortality in Santiago relative to its synthetic counterfactual within the same comuna and week. Columns (1) and (2) use exposure from all 11 contracts. Columns (3)–(6) come from specifications that separately include exposure generated by contracts beginning in 2023 and 2025. The good-pre-fit sample restricts attention to the 37 treated comunas with a 2015–2022 mortality-path RMSE no greater than 0.05, a criterion fixed using pre-treatment outcomes only. Partial calendar weeks containing a formal contract start are omitted. Standard errors two-way clustered by treated comuna and calendar week are reported in parentheses. The number of observations shown corresponds to the mortality outcomes; the non-transport external-death placebo uses 19,034 observations in the full sample and 16,794 in the good-pre-fit sample because PPML removes a small number of perfect-fit observations. Coefficients are semi-elasticities. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.